The Coupling Integrals of Polynomial Cubes in Closed Form

Exact for Elastic Waves, and Accurate in Double Precision

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Abstract

Volume-integral methods for elastic waves divide a heterogeneous medium into cubic cells and couple every pair of cells through the integral of the elastodynamic Green's tensor against the polynomials that the two cells carry. We obtain these coupling blocks in closed form, exact for elastic waves and accurate to round-off in double precision. The expansion about the two cell centres that couples separated scatterers is assured to converge only when the cells' circumscribing spheres do not overlap, a sufficient condition that every pair of touching cells violates, and at face and edge contact it fails. Evaluated in closed form, the coupling integrals need no such condition. They are singular for cells that touch, and their closed forms are a power series in the wavenumber whose coefficients are universal numbers, independent of the frequency, the cell size and the medium, each built from elementary master integrals. Written in powers of the coordinates about the singular point, as the master integrals naturally are, they inherit the classical ill-conditioning of power moments and lose four to five digits in double precision: against a forty-digit reference, the block of a cell carrying a degree-two field and its corner neighbour is then wrong by 1.7×10^{-11}, where quadrature reaches 1.9×10^{-15}. We evaluate the same exact integrals in a Legendre basis instead, accurate to round-off: the pieces of the integration domain away from the singular point are expanded in Legendre polynomials of their own coordinates, whose moments follow from a banded relation posed as a truncated boundary-value problem, with no transcendental function evaluated; the pieces at the singular point are split into Duffy pyramids. The touching blocks are then accurate to 9.7×10^{-16} to 4.3×10^{-15} for fields of degree one and two. Cells that do not touch are coupled by the same series with coefficients computed once per offset, and distant cells also by a two-centre multipole series where it converges. Every block at every frequency is then a weighted sum of stored numbers, and over 256 frequencies the series costs 1.5 to 1.9 times less than quadrature, its one-off cost included. Because the coefficients are exact, the response of a whole cell model can be expanded in powers of the frequency, each power by one solve with the static operator. For a graded sphere modelled by Legendre cells of degree two, every coefficient of the far field from $(ka)^2$ to $(ka)^7$ is accurate to 8×10^{-8} to 2×10^{-7} at 12 cells across, with apparent orders of 6.2 to 7.5 in the cell size. Cells of degree one converge at fourth order, the apparent orders reaching 3.96 to 4.00 at 20 cells across, and at 12 cells are 43 to 8100 times less accurate. Both reproduce the exact zero of the $(ka)^3$ coefficient to round-off. Summed at each frequency of a sweep, the same coefficients give the τ–p section of the sphere on a plane at 1.5 radii from its centre, evanescent waves included, with no far-field

approximation: the largest difference from the exact section, relative to its peak, is 1.7×10^{-6} with cells of degree two and 1.2×10^{-4} with cells of degree one.

1 Introduction

A volume-integral method for elastic scattering represents a heterogeneous medium by cells, here cubes, each carrying a polynomial field, and couples every pair of cells through the integral of the elastodynamic Green's tensor against the two cells' polynomials. Higher-order cells are worth having only if these coupling integrals are accurate: a cell whose field converges at fourth order in the cell size [1] needs couplings correct to a matching tolerance, and in practice to round-off, because errors in the touching blocks are not reduced by refinement. For cells that touch, the integrand is singular. Quadrature with Duffy's transformation [2] at the singular vertex handles it, but it must be repeated for every frequency, and its accuracy has to be established case by case. Closed forms avoid both: the singular integrals are a power series in the wavenumber whose coefficients are universal numbers, independent of the frequency, the cell size and the medium, each a finite sum of elementary master integrals reduced from volume to faces to edges.

The cells here fill space. That choice is what makes the touching integrals unavoidable. Methods that leave gaps between their scatterers couple them through expansions about each scatterer's own centre, which are assured to converge when the scatterers are far enough apart: the circumscribing spheres must not overlap, a sufficient condition for the convergence of the translation series [3] that superposition T-matrix methods adopt [4], and point-scatterer models of the Foldy type approximate each small obstacle by a point scatterer [5, 6]. In a space-filling tessellation every cell touches 26 neighbours, and the condition fails for all of them; the expansion itself fails at face and edge contact (§3). The interaction between neighbours, which in a continuous medium is the coupling of adjacent material, therefore needs the integrals themselves. Evaluated in closed form they need no condition on the geometry, so the condition, sufficient for the expansion, is not necessary for coupling the cells. This paper follows for that interaction the programme that Nestor [7] carried out for a single cell, whose single-site T-matrix was obtained there in closed form at every power of the frequency: here the interaction between cells is obtained the same way. With both in closed form and accurate to round-off, the errors that Nestor [1] measures are those of the discretisation alone, with no quadrature error mixed into them. The closest prior work we know is the analytical integration of the Green's tensor over cuboids for the electromagnetic discrete dipole approximation [8], which neglects terms of fourth order in kd; the integrals here are elastic, exact at every order through a series in the wavenumber, and over cells that carry polynomials.

This paper is about evaluating those closed forms to round-off. Written in powers of the coordinates about the singular point, as the master integrals naturally are, they inherit the known ill-conditioning of power moments [9], and in double precision they are much less accurate than quadrature (§4). We measure by how much against a reference computed at forty digits by a route that shares nothing with them (§6.1), locate where the digits go, and evaluate the same exact integrals in a Legendre basis that is accurate to round-off (§5). The loss is a known phenomenon in another setting: in boundary-element methods the analytic integration of layer

potentials over flat elements loses accuracy away from the singularity and with the order of the element, and the remedy adopted there is to switch to quadrature beyond some distance [10, 11]. Here the closed forms are kept everywhere and made stable instead, by changing the basis in which they are assembled. The underlying facts are classical: moments in powers of a variable are progressively ill-conditioned while moments in orthogonal polynomials are well conditioned [9], and a recurrence whose wanted solution is dominated by another solution cannot be marched, but can be solved as a truncated boundary-value problem [12, 13]. What is new is the relation that gives the Legendre moments of a power of the distance over a box in any dimension, and the decomposition that brings every piece of a touching coupling block within its reach.

The contributions are the following.

1. For cells that touch, the two-centre expansion that couples separated scatterers diverges at face contact, stalls at edge contact and converges only algebraically at corner contact, whatever the precision of the arithmetic; and no subdivision of the cells can restore it (§3).

2. The touching blocks are a power series in the wavenumber with universal coefficients, each a finite sum of elementary master integrals; assembled in powers about the singular point they lose four to five digits, and the loss is located in the pieces away from the singular point and traced to the power basis (§4).

3. A banded relation for the Legendre moments of a power of the distance over a box in any dimension, solved as a truncated boundary-value problem, together with a Duffy decomposition of the pieces at the singular point, evaluates the same exact integrals to round-off (§5); the blocks agree with an independent forty-digit reference to 10^{-15} (§6).

4. Cells that do not touch are coupled to round-off by the same series with coefficients computed once per offset, and the whole operator over a frequency sweep costs less than quadrature at every frequency (§7).

5. Because the coefficients are exact and independent of the frequency, the response of a cell model can be expanded in powers of the frequency, each power by one solve; for a graded sphere every power is compared with the exact one, and summed over a sweep the same coefficients give a τ–p section in the near field (§8).

The paper is organised as follows. Section 2 defines the blocks, their separation form and their series in the wavenumber. Section 3 shows why the touching blocks cannot be had from the two-centre expansion. Section 4 gives the closed forms and their conditioning in the power basis, §5 the stable evaluation in a Legendre basis, and §6 the forty-digit reference and the accuracy of the blocks. Section 7 treats cells that do not touch and the cost of a frequency sweep, §8 the graded sphere, and §9 the relation to quadrature, to boundary-element practice and to the fast multipole method.

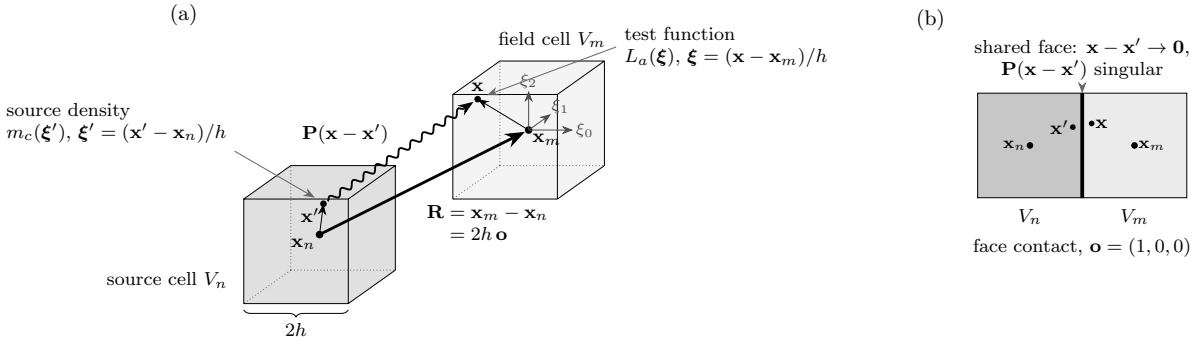


Figure 1. The geometry of the coupling block, Eq. (1). (a) A source cell V_n and a field cell V_m, cubes of side $2h$ with centres $\mathbf{x}_n$ and $\mathbf{x}_m$ and integer offset $\mathbf{o} = \mathbf{R}/2h$, here $(2, 0, 1)$. Each point is located by its cell's local coordinates, $\boldsymbol{\xi}'$ and $\boldsymbol{\xi}$ in $[-1, 1]^3$ (the field cell's axes shown). The source density $m_c(\boldsymbol{\xi}')$ at a source point $\mathbf{x}'$ radiates through the point propagator $\mathbf{P}(\mathbf{x} - \mathbf{x}')$ to a field point $\mathbf{x}$, where it is weighted by the test function $L_a(\boldsymbol{\xi})$; the block is this product integrated over both cells. (b) A section through two cells in face contact: on the shared face (thick) the field and source points coincide, so the integrand is singular there. For edge and corner contact the cells share an edge or a vertex, and the self block has $\mathbf{x} = \mathbf{x}'$ throughout the cell.

2 The coupling blocks of polynomial cubes

Cells and fields. A cell is a cube $[-h, h]^3$ of half-width h centred at $\mathbf{x}_c$, with local coordinates $\boldsymbol{\xi} = (\mathbf{x} - \mathbf{x}_c)/h \in [-1, 1]^3$. It carries the nine-component state, the displacement and an independent engineering strain, as a polynomial in $\boldsymbol{\xi}$: its Legendre moments of degree at most one (four functions: the degree-one cell) or two (ten: the degree-two cell) [7]. The source density, the contrast times the state, is a polynomial in the source cell's coordinates $\boldsymbol{\xi}'$, written in the monomials $m_c(\boldsymbol{\xi}') = \xi_0'^{e_0}\xi_1'^{e_1}\xi_2'^{e_2}$ of degree at most two or four (10 or 35 of them). With $\mathbf{P}$ the 9×9 point propagator, built from the Green's tensor and up to two of its derivatives, the cells m and n couple through the blocks

$$\mathbf{K}_{ac}(\mathbf{R}) = \int_{V_m}\int_{V_n} L_a(\boldsymbol{\xi})\,\mathbf{P}(\mathbf{x} - \mathbf{x}')\,m_c(\boldsymbol{\xi}')\,\mathrm{d}^3x\,\mathrm{d}^3x', \qquad \mathbf{R} = \mathbf{x}_m - \mathbf{x}_n, \tag{1}$$

L_a a test function of the field cell (Figure 1). They are computed with test monomials and recombined into the orthogonal L_a. The integer offset between the cells is $\mathbf{o} = \mathbf{R}/2h$; the cells touch when $\max_i |o_i| \le 1$, and the self cell and its 26 neighbours fall into four geometries up to the symmetries of the cube: self, face, edge and corner contact.

The s-form. With $\mathbf{x} = \mathbf{x}_m + \mathbf{u}$ and $\mathbf{x}' = \mathbf{x}_n + \mathbf{u}'$ the kernel sees $\mathbf{R} + \mathbf{s}$, $\mathbf{s} = \mathbf{u} - \mathbf{u}' \in [-2h, 2h]^3$, and

$$\mathbf{K}_{ac}(\mathbf{R}) = \int W_{ac}(\mathbf{s})\,\mathbf{P}(\mathbf{R} + \mathbf{s})\,\mathrm{d}^3s, \qquad W_{ac}(\mathbf{s}) = \prod_i w_{e_{a,i}e_{c,i}}(s_i), \tag{2}$$

with the one-dimensional cross-correlation $w_{\alpha\beta}(s) = h\int v^\alpha (v - s/h)^\beta\,\mathrm{d}v$ over $v \in [\max(-1, s/h - 1), \min(1, s/h + 1)]$. Each factor is a polynomial on $[-2h, 0]$ and on $[0, 2h]$, continuous, zero at $\pm 2h$, and kinked at 0 and $\pm 2h$. In units of h, $\sigma = s/h$, the domain of Eq. (2) is cut into 8 boxes on which W_{ac} is a polynomial, whose corners lie on the breakpoints $\sigma_i \in \{-2, 0, 2\}$. The kernel is

singular at $\mathbf{s} = -\mathbf{R}$, that is at $\boldsymbol{\sigma}^* = -2\mathbf{o}$, which for touching cells is always a breakpoint: a vertex of some of the boxes, and at a distance of at least 2 (in units of h) from the others.

The series in the wavenumber. With $g_k = \mathrm{e}^{\mathrm{i}kr}/r = \sum_{t\geq 0}(\mathrm{i}k)^t r^{t-1}/t!$, the Green's tensor is a sum over powers r^m, $m = t-1 \geq -1$, of two families,

$$\mathbf{G}_{ij} = \frac{1}{4\pi\mu}\sum_{m\geq -1}\Big[c_1(m)\,\delta_{ij}\,r^m + c_2(m)\,\partial_i\partial_j r^m\Big], \qquad c_1 = \frac{(\mathrm{i}k_S)^{m+1}}{(m+1)!}, \quad c_2 = \frac{\mathrm{i}^{m+1}\big(k_S^{m+1}-k_P^{m+1}\big)}{k_S^2\,(m+1)!}, \tag{3}$$

with μ the shear modulus and k_P, k_S the compressional and shear wavenumbers of the background. The static (Kelvin) tensor is c_1 at $m=-1$ and c_2 at $m=1$. Since $\mathbf{P}$ adds up to two derivatives, every entry of every block is a combination, with 9×9 coefficient tables fixed by the propagator's structure, of the scalar moments

$$\int_V\int_{V'} L_a(\boldsymbol{\xi})\,\partial^{\boldsymbol{\iota}} r^m\, m_c(\boldsymbol{\xi}')\,\mathrm{d}^3x\,\mathrm{d}^3x' = h^{\,6+m-|\boldsymbol{\iota}|}\,U_{ac}(m,\boldsymbol{\iota};\mathbf{o}), \qquad |\boldsymbol{\iota}|\leq 4, \tag{4}$$

where $\boldsymbol{\iota}$ is the multi-index of derivatives with respect to $\mathbf{x}$. The U are pure numbers: they depend on neither the frequency, the cell size nor the medium. For even m, r^m is a polynomial and the moment is an integral of polynomials over the boxes, exact by a Gauss rule of sufficient order. The odd m carry the singularity, and they are the subject of this paper.

Moving derivatives. For $m=-1$ and $m=1$ with enough derivatives the kernel $\partial^{\boldsymbol{\iota}} r^m$ is a distribution at the singular point. Of the derivatives in $\boldsymbol{\iota}$, $k = \max\big(0, |\boldsymbol{\iota}| - \min(m+1, \frac{m+3}{2})\big) \leq 2$ are moved onto W: $\int W\,\partial^{\boldsymbol{\iota}} f = (-1)^k \int (\partial^{\text{moved}} W)\,\partial^{\text{rest}} f$. What remains, $\partial^{\text{rest}} r^m$, is at most $1/r$ singular, and it is a sum of monomials times odd powers of the distance,

$$\partial^{\text{rest}} r^m = \sum_j \gamma_j\,\mathbf{t}^{\boldsymbol{\alpha}_j}\,|\mathbf{t}|^{n_j}, \qquad \partial_i\big(\mathbf{t}^{\boldsymbol{\alpha}}|\mathbf{t}|^n\big) = \alpha_i\,\mathbf{t}^{\boldsymbol{\alpha}-\mathbf{e}_i}|\mathbf{t}|^n + n\,\mathbf{t}^{\boldsymbol{\alpha}+\mathbf{e}_i}|\mathbf{t}|^{n-2}, \tag{5}$$

in the separation $\mathbf{t} = \boldsymbol{\sigma} - \boldsymbol{\sigma}^*$ measured from the singular point, with $|\boldsymbol{\alpha}_j| + n_j = m - |\text{rest}|$ for every j. Two moved derivatives on one axis turn the kinks of w into plane deltas $J_b\,\delta(\sigma - b)$, $b\in\{-2,0,2\}$, J_b the jump of w' at b. Each term of a moment is therefore an integral of a polynomial in $\mathbf{t}$ times $\mathbf{t}^{\boldsymbol{\alpha}}|\mathbf{t}|^n$ over a box of side 2 whose corners lie on the breakpoints, or over a face of one when a plane delta fixes a coordinate. This is the object every route below evaluates; they differ only in how.

3 The two-centre expansion for cells that touch

The two-centre expansion. The block (2) depends on the two cells only through the separation $\mathbf{R}+\mathbf{s}$ of a field point and a source point. Expanding the propagator about the separation of the centres, $\mathbf{P}(\mathbf{R}+\mathbf{s}) = \sum_{\boldsymbol{\gamma}} \mathbf{s}^{\boldsymbol{\gamma}}\partial^{\boldsymbol{\gamma}}\mathbf{P}(\mathbf{R})/\boldsymbol{\gamma}!$, and integrating term by term over the

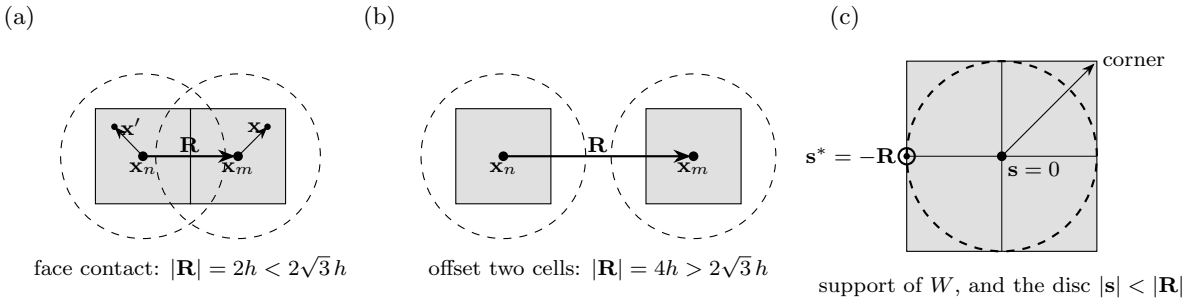


Figure 2. The two-centre expansion, in a section through the cell centres. (a) Two cells in face contact, centres $\mathbf{x}_n$ and $\mathbf{x}_m$ a distance $|\mathbf{R}| = 2h$ apart, with a source point $\mathbf{x}'$ and a field point $\mathbf{x}$; the dashed circles are the sections of the circumscribing spheres, of radius $\sqrt{3}\,h$, which overlap in the moon region [14, ch. 3]. (b) Cells two widths apart, $|\mathbf{R}| = 4h > 2\sqrt{3}\,h$: the spheres are disjoint. (c) The separation $\mathbf{s} = (\mathbf{x} - \mathbf{x}_m) - (\mathbf{x}' - \mathbf{x}_n)$ for face contact: the shaded square is the support of the pair's cross-correlation W (four of its eight pieces in this section), the dashed circle the disc $|\mathbf{s}| < |\mathbf{R}|$ on which the Taylor series of $\mathbf{P}(\mathbf{R} + \mathbf{s})$ about $\mathbf{s} = 0$ converges, and the ringed point the singular point $\mathbf{s}^* = -\mathbf{R}$, on the boundary of the support. The support reaches beyond the disc, to $|\mathbf{s}| = 2\sqrt{3}\,h$ at its corners in three dimensions.

two cells gives

$$\mathbf{K}_{ac}(\mathbf{R}) = \sum_{\gamma} \frac{h^{\,6+|\gamma|}}{\gamma!}\, \mu_{ac}^{\gamma}\, \partial^{\gamma}\mathbf{P}(\mathbf{R}), \qquad \mu_{ac}^{\gamma} = \iint \boldsymbol{\xi}^{\mathbf{e}_a} (\boldsymbol{\xi} - \boldsymbol{\xi}')^{\gamma}\, \boldsymbol{\xi}'^{\mathbf{e}_c}\, \mathrm{d}^3\xi\, \mathrm{d}^3\xi', \tag{6}$$

with the multi-index $\boldsymbol{\gamma}$ running over all orders. The pair moments μ_{ac}^{γ} are products of one-dimensional integrals of monomials, rational numbers, and the separation enters only through the derivatives of the propagator at one point, which are elementary. No singular integral appears: the two cells are replaced by their multipole moments about their own centres (Figure 2(a)). Equation (6) is the Cartesian counterpart of the translation addition theorem by which superposition T-matrix methods couple scatterers, each represented about its own origin [4], and it is the series used for distant cells in §7.

The condition, and why every neighbour violates it. The Taylor series of $\mathbf{P}(\mathbf{R} + \mathbf{s})$ about $\mathbf{s} = 0$ converges for $|\mathbf{s}| < |\mathbf{R}|$, the distance from the expansion point to the singular point $\mathbf{s}^* = -\mathbf{R}$. The separation ranges over the support of W, $|s_i| \le 2h$, as far as $|\mathbf{s}| = 2\sqrt{3}\,h$, the sum of the radii of the two cells' circumscribing spheres. The series is therefore assured to converge only when $2\sqrt{3}\,h < |\mathbf{R}|$, that is when the circumscribing spheres do not overlap (Figure 2(b)); this is the sufficient condition for the convergence of the translation (addition) theorems of the Laplace and Helmholtz kernels [3]. Where the spheres of touching cells overlap, the expansion about one centre must reach into the neighbouring cell. Multiple scattering theory in solids, whose space-filling cells meet the same difficulty, calls this region of overlap the moon region [14, ch. 3]. Superposition T-matrix methods state the same condition: the smallest circumscribing spheres of the components must not overlap [4, §5.9], and each particle's circumscribing sphere must not intersect its neighbours [15]. The condition is sufficient, not necessary: for adjacent spheroids, results converge well beyond it, although extreme overlap needed quadruple precision [16]. A space-filling tessellation violates it for each of a cell's 26 neighbours (Figure 3), with the

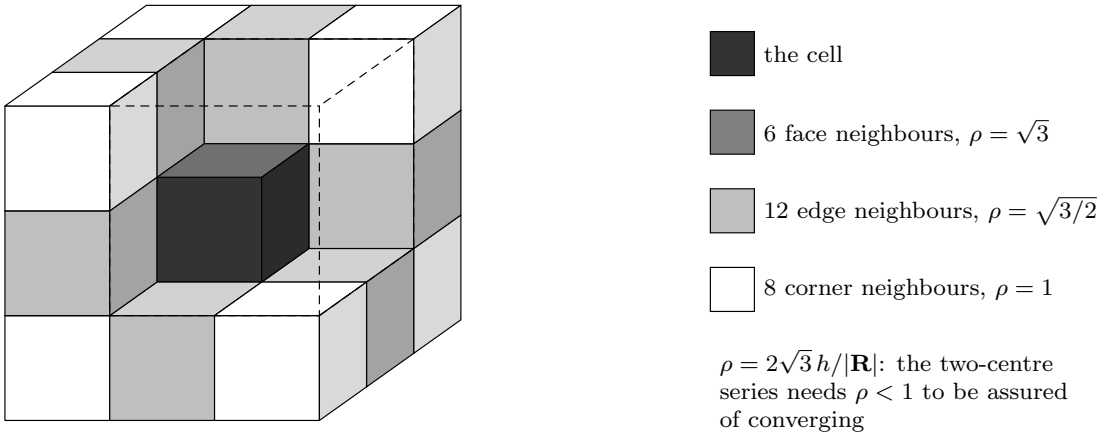


Figure 3. A cell of a space-filling tessellation and its 26 neighbours, packed face to face as in the medium. The seven neighbours in front of, above and to the right of the cell are cut away (dashed outline) so that the cell (darkest) shows. The neighbours touch it across a face (6), an edge (12) or a corner (8); for each the ratio $\rho = 2\sqrt{3}\,h/|\mathbf{R}|$ of the two circumscribing radii to the distance of the centres is at least one, so none satisfies the condition under which the two-centre series is assured to converge (Table 1).

ratio $\rho = 2\sqrt{3}\,h/|\mathbf{R}|$ equal to $\sqrt{3}$, $\sqrt{3/2}$ and 1 at face, edge and corner contact, and for touching cells the singular point lies on the boundary of the support (Figure 2(c)). The condition being only sufficient, whether the series converges there anyway can be settled only by computing it against an exact value. The closed forms of this paper supply one at every contact, and need no condition on the geometry themselves.

The test. We test the series on the coupling moment of the kernel $1/r$, $U_{ac}(\mathbf{o}) = \int\int \boldsymbol{\xi}^{\mathbf{e}_a}\,|\mathbf{x} - \mathbf{x}'|^{-1}\,\boldsymbol{\xi}'^{\mathbf{e}_c}$, for cells of half-width $h = 1$ carrying a degree-one field (four test monomials and ten source monomials). The kernel $1/r$ is the static singular part that every family of the propagator shares (the term $m = -1$ of Eq. (3)), so its behaviour is the behaviour of the series for every block. The exact value is, for cells that touch, the stable closed form of §5, accurate to round-off against the forty-digit reference of §6.1; for cells apart, Gauss rules of 32 points per axis on each of the eight pieces of the s-form, which reach round-off because the integrand is then smooth. The series (6) is summed in 60-digit arithmetic, with the pair moments exact as rationals and the derivatives of $1/r$ at $\mathbf{R}$ from their Hermite-type expansion with integer coefficients, so that divergence cannot be mistaken for rounding. The partial sums S_N, over all $|\boldsymbol{\gamma}| \le N$, are compared with the exact value; the same sums in double precision give the condition number $\kappa_N = \sum|\text{terms}|/|S_N|$ over the entries larger than 10^{-6} of the largest.

The result. The series fails for touching cells (Table 1). At face contact it diverges: at every fourth order from 0 to 48 its error is at least 10^{-2}, the smallest at $N = 4$ and 8, and from $N = 12$ it grows, to 1.2×10^{3} at $N = 48$. At edge contact it reaches 1.3×10^{-5} at $N = 24$ and then grows again, to 6.6×10^{-5} at $N = 48$, without converging. At corner contact, where the circumscribing spheres just touch, it converges, but only algebraically, to 5×10^{-8} at $N = 48$. Cells that do not touch converge geometrically, the faster the smaller ρ, as the condition predicts; at the offset $(3, 0, 0)$ the series reaches round-off by $N = 32$. Precision does not change this: at the three

offset $\mathbf{o}$	ρ	$N = 8$	$N = 16$	$N = 32$	$N = 48$	κ_{48}
$(1,0,0)$, face	1.73	1.0×10^{-2}	1.5×10^{-2}	1.5	1.2×10^{3}	2.2×10^{12}
$(1,1,0)$, edge	1.22	3.5×10^{-4}	5.2×10^{-5}	3.1×10^{-5}	6.6×10^{-5}	1.0×10^{6}
$(1,1,1)$, corner	1.00	8.8×10^{-5}	6.3×10^{-6}	3.4×10^{-7}	5.4×10^{-8}	4.6×10^{2}
$(2,0,0)$	0.87	1.2×10^{-5}	1.6×10^{-7}	3.2×10^{-10}	2.6×10^{-12}	3.5
$(2,1,1)$	0.71	2.1×10^{-6}	4.6×10^{-9}	3.2×10^{-13}	7.4×10^{-16}	6.5
$(3,0,0)$	0.58	3.7×10^{-7}	1.6×10^{-10}	1.3×10^{-15}	1.0×10^{-15}	1.4

Table 1. The two-centre series (6) for the kernel $1/r$: error of the partial sum S_N, summed in 60-digit arithmetic, relative to the largest entry of the exact moment, and the condition number of the sum in double precision at $N = 48$. $\rho = 2\sqrt{3}\,h/|\mathbf{R}|$ is the ratio of the two circumscribing radii to the distance of the centres.

touching geometries the 60-digit and double-precision sums agree to every digit shown, so the failure is the series', not the arithmetic's, and the remedy that rescued adjacent spheroids, more digits, does nothing here. The condition number of the sum tells the same story from the other side: κ_{48} is 2×10^{12} at face contact and of order one for the separated cells.

Subdivision does not rescue it. For cells that do not touch, the series can be made to converge faster by cutting the support into boxes and expanding about each box's own centre (§7.2). For cells that touch it cannot: a box whose closure holds the singular point $\mathbf{s}^*$ has its centre within one half-diagonal of it, so its own ratio is at least one, however finely the support is cut, and such a box always exists because $\mathbf{s}^*$ lies on the boundary of the support. The interaction of a cell with its neighbours, which carries the singular part of the coupling in a space-filling medium, is therefore out of reach of the expansion that methods leaving gaps between their scatterers rely on. In a homogeneous reference medium it is also where the coupling is strongest. In a stratified one, reflections and guided waves couple distant cells as strongly, but the singular part of the Green's tensor is still that of a homogeneous medium with the local properties, so the touching blocks need these integrals whatever the reference; next to an interface the reflected part is singular as well, and is treated separately [17]. Two statements should be kept apart. For the series, the condition is not necessary but it is not idle either: the series converges at corner contact, where the spheres just touch, and fails at face and edge contact however its terms are evaluated, since here the pair moments are exact and the sums carried to 60 digits. For the coupling, the condition is not necessary at all: the closed forms of the rest of this paper give the block at every contact to round-off, against the forty-digit reference at edge and corner contact and against quadrature at face contact (§6). The condition belongs to the expansion, not to the problem.

4 Closed forms, and the conditioning of power moments

The closed forms reduce every moment to master integrals of monomials against a power of the distance, anchored at the singular point. Written so, a moment is a sum over power moments of the polynomial weight, and that power (ordinary) moments are ill-conditioned while moments of orthogonal polynomials (modified moments) are well conditioned is classical [9]. This section

measures what the power basis costs for the coupling integrals, and where the cost arises, so that the evaluation of §5 can be judged against it.

Measuring the conditioning. Every evaluation below ends in a sum, $S = \sum_n a_n$, of terms computed to a relative accuracy near the unit round-off ε. Its rounding error is then at most about $\varepsilon \sum_n |a_n|$, so that the sum loses about $\log_{10} \kappa$ significant digits, with

$$\kappa = \frac{\sum_n |a_n|}{\left|\sum_n a_n\right|} \tag{7}$$

its condition number [18]; $\kappa = 10^8$ leaves about eight good digits in double precision. An integral has a condition number of its own, $\kappa_\int(f) = \int |f| \,/\, \left|\int f\right|$, and it is the floor below which no method that works from the values of the integrand can be expected to go. If the integrand is perturbed by δf with $|\delta f| \le \varepsilon |f|$ at every point, then $|\int \delta f| \le \varepsilon \int |f|$, so the relative change of the integral is at most $\varepsilon\, \kappa_\int(f)$, and the perturbation $\delta f = \varepsilon |f| \operatorname{sgn} \int f$ attains it. A Gauss rule with positive weights computes $\sum_i w_i f(\mathbf{x}_i)$, whose condition number tends to $\kappa_\int(f)$ as the rule converges, because the same weights integrate $|f|$; quadrature therefore loses about as many digits as the problem itself forces. A closed form instead writes the integral as a sum in a basis of its own choosing, and its condition number κ can be far larger than $\kappa_\int$: that excess, and nothing else, is what the choice of basis decides. The rest of this section gives the closed forms and measures that excess for the power basis.

The master integrals. The integrand $\mathbf{t}^a |\mathbf{t}|^n$ is homogeneous about the singular point, and three master integrals anchored there are elementary:

$$I[p,q,r;a,b,c;n] = \int_0^a \int_0^b \int_0^c x^p y^q z^r \,(x^2+y^2+z^2)^{n/2}\, \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x, \tag{8}$$

$$J[q,r;a;b,c;n] = \int_0^b \int_0^c y^q z^r \,(a^2+y^2+z^2)^{n/2}\, \mathrm{d}z\, \mathrm{d}y, \tag{9}$$

$$L[r;A;c;n] = \int_0^c z^r \,(A^2+z^2)^{n/2}\, \mathrm{d}z, \tag{10}$$

with n odd. Euler's identity for a homogeneous function and the divergence theorem reduce the box to faces, the faces through the singular point carrying no flux; on a face, the derivative of $y^{q-1}(a^2+y^2+z^2)^{(n+2)/2}$ lowers the power of y by two and leaves edge integrals; on an edge the same step ends in a power, a square root or an inverse hyperbolic sine. One transcendental base case is left on the faces, $J[0,0;a;b,c;-3] = a^{-1} \arctan\big(bc/(a\sqrt{a^2+b^2+c^2})\big)$, $1/a$ times the solid angle the rectangle subtends at the singular point. The recursions divide by small integers only, and they were evaluated at forty digits. To use them for a box that is not anchored at the singular point, two steps are needed. On each axis an integral from l to u is the difference of two integrals from the origin (the singular point), so a box at a distance becomes a signed sum of eight anchored boxes of sides 2 and 4. And the polynomial weight is expanded in powers of $\mathbf{t}$, so that each term is a master integral. Evaluated in this power basis, the master integrals are rounded to double precision and contracted with the expanded weights in double precision.

What the power basis costs. Against the forty-digit reference of §6.1, the static coupling block assembled this way is wrong by 2.3×10^{-13} (degree-one field, corner contact), 9.4×10^{-14} (degree one, edge), 1.7×10^{-11} (degree two, corner) and 5.5×10^{-12} (degree two, edge), relative error in the Frobenius norm over all entries of the block. Quadrature of the same block (Duffy pyramids at the singular vertex and tensor Gauss rules of 20 points per axis on the pieces) is accurate to 1.1×10^{-15}, 1.8×10^{-15}, 1.9×10^{-15} and 2.7×10^{-15}. In the power basis the closed forms, exact as they are, are less accurate than quadrature by two orders of magnitude for a degree-one field and four for a degree-two one.

Where the cost arises. Each block is a sum over the pieces of W. Taking one term, $m = 1$ with $\iota = (0, 2, 2)$, at corner contact with a degree-one field, and evaluating the same algorithm once in double precision and once with every step at forty digits, the difference piece by piece is 2.1×10^{-12} (relative to the largest entry of the term) on the piece farthest from the singular point, the box $\mathbf{t} \in [2, 4]^3$, and at most 1.2×10^{-15} on the piece that has the singular point as a vertex. The loss is in the pieces away from the singular point.

The cause. On such a piece two cancellations act together: the inclusion-exclusion of anchored boxes, and the power basis about a point outside the piece. They can be separated. On the far piece of the corner contact, for the same term, the integral was computed four ways, each compared with the same piece at forty digits with exact rational weights:

A the power basis about the singular point: anchored master integrals rounded to double precision, inclusion-exclusion and contraction in double precision. Error 2.6×10^{-13}.

B the moments of the far box itself, $\int_{\text{box}} \mathbf{t}^{\mathbf{e}} |\mathbf{t}|^n$, exact (inclusion-exclusion at forty digits) and then rounded; the contraction with the power-basis weights in double precision. Error 6.7×10^{-14}.

C as B, with the contraction at forty digits too, the weights still rounded to double precision. Error 8.6×10^{-15}.

D the weight re-expanded about the piece's own corner, in the local variable $\boldsymbol{\tau} = \mathbf{t} - \mathbf{l} \in [0, 2]^3$, with the local moments $\int \boldsymbol{\tau}^{\mathbf{j}} \mathbf{t}^{\alpha} |\mathbf{t}|^n$ exact and rounded, and a double-precision contraction. Error 6.8×10^{-16}.

Inclusion-exclusion accounts for a factor of four (A against B). The rest is the power basis about the singular point: the moments reach 2×10^{19} against a piece of order one, so that the rounding of the weights alone (C) costs three digits. A basis local to the piece removes the loss entirely (D). At degree two the local power basis is itself not quite enough: with exact moments the far piece is accurate to 4.9×10^{-15} in local powers and to 1.9×10^{-16} in Legendre polynomials of the local coordinate, and the piece at the singular point to 4.0×10^{-15} and 2.4×10^{-16}. The basis the evaluation needs is therefore the Legendre basis of each piece, as the classical contrast between power and modified moments leads one to expect [9].

Quadrature on the smooth pieces is not enough. A natural alternative keeps the closed forms only on the pieces that hold the singular point and integrates the others, which are smooth, by Gauss rules. It cures the degree-one field (9.1×10^{-16} and 1.6×10^{-15} at corner and edge) but not the degree-two one (5.9×10^{-15} and 1.1×10^{-14}, insensitive to the Gauss order between 12 and 20), because at degree two the pieces at the singular point lose digits too, to the power basis on $[0, 2]$. The same switch is the remedy adopted for analytic boundary-element integrals [10, 11]; here it does not reach round-off, and it gives up the frequency-independent coefficients only on part of the domain while keeping the ill-conditioned basis on the rest.

5 A stable evaluation

5.1 Marching the recurrences

The reductions of §4 can be re-anchored at the corner of each piece, so that the weights are local. In one dimension the edge moments $E[r] = \int_l^{l+L} u^r (A^2 + t^2)^{n/2} \mathrm{d}t$, with $t = u + l$, satisfy at fixed n the three-term recurrence

$$(r + n + 2)\, E[r + 1] + l\,(2r + n + 2)\, E[r] + r\,(A^2 + l^2)\, E[r - 1] = \left[u^r (A^2 + t^2)^{(n+2)/2}\right]_{t=l}^{t=l+L}, \qquad (11)$$

from $\frac{\mathrm{d}}{\mathrm{d}t}\left(u^r (A^2 + t^2)^{(n+2)/2}\right)$ with $(A^2 + t^2) = (A^2 + l^2) + 2lu + u^2$. Its homogeneous solutions grow like ρ^r, with $\rho = \sqrt{A^2 + l^2}$ the distance from the singular point to the near end of the edge, while the wanted moments grow like L^r. Away from the singular point ($\rho > L$) the wanted solution is the minimal one, forward recursion amplifies the rounding errors, and backward recursion from zero values at a high degree converges to it, the error falling like $(L/\rho)^{R-r}$ [12, 13]. Against forty-digit quadrature, forward recursion loses up to 10^{-7} by degree 20 when $\rho/L = 2.45$, and backward recursion is at round-off (2×10^{-16}) for every ρ/L tested, from 1.41 to 2.45, including $n = -3$, where the forward recursion fails at once on a zero divisor. Backward recursion needs no transcendental starting value, only the end terms of Eq. (11). For $\rho \le L$, near the singular point, forward recursion is the stable direction.

In two dimensions the same identity along one axis acquires cross terms in the other, since $(c^2 + t_y^2 + t_z^2)$ involves both, and running it backwards requires a back-substitution in the other index at every level. This works to round-off (at most 10^{-15}) on every face tested whose near corner is at least 1.73 side lengths from the singular point, but it diverges, to errors of 10^2 to 10^{97}, on the faces at $\sqrt{2}$ side lengths; forward recursion there reaches only 10^{-9} to 2×10^{-8} at degree 8, and 4×10^{-13} to 3×10^{-11} at a distance of one side length. There is a band of geometries next to the singular point in which neither direction is stable, and the faces of the pieces adjacent to the singular point fall in it. Marching does not give a complete scheme.

5.2 Legendre moments by a boundary-value relation

On the unit box $\prod_j [l_j, l_j + 1]$ in $d = 1$, 2 or 3 dimensions, lifted by $c^2 \geq 0$ so that $S^2 = c^2 + \sum_j t_j^2$, define the Legendre moments

$$\Lambda[\mathbf{k}] = \int_{\text{box}} \prod_j P_{k_j}(x_j)\, S^m \, \mathrm{d}^d t, \qquad x_j = 2(t_j - l_j) - 1 \in [-1, 1], \tag{12}$$

for boxes whose closure does not hold the singular point ($S > 0$ on the box). Along an anchor axis i, the Legendre identity $(2k+1)P_k = P'_{k+1} - P'_{k-1}$ and an integration by parts in x_i, applied to $g = S^{m+2} = (c^2 + \sum_j t_j^2)\, S^m$, whose derivative is $\partial g / \partial x_i = \frac{m+2}{2}\, t_i S^m$, give

$$(2k_i + 1) \int P_{\mathbf{k}} \Big(c^2 + \sum_j t_j^2\Big) S^m + \frac{m+2}{2} \int \big(P_{k_i+1} - P_{k_i-1}\big)(x_i)\, t_i\, P_{\hat{\mathbf{k}}}\, S^m = 0, \qquad k_i \geq 1, \tag{13}$$

where $P_{\mathbf{k}} = \prod_j P_{k_j}(x_j)$ and $\hat{\mathbf{k}}$ collects the other axes. The boundary terms vanish because $P_{k_i+1} - P_{k_i-1}$ is zero at $x_i = \pm 1$. For $k_i = 0$, with $P_0 = P'_1$,

$$\int P_{\mathbf{k}} \Big(c^2 + \sum_j t_j^2\Big) S^m + \frac{m+2}{2} \int P_1(x_i)\, t_i\, P_{\hat{\mathbf{k}}}\, S^m = \tfrac{1}{2}\Big(\Lambda_{d-1}\big[\hat{\mathbf{k}}; c^2 + (l_i+1)^2\big] + \Lambda_{d-1}\big[\hat{\mathbf{k}}; c^2 + l_i^2\big]\Big), \tag{14}$$

the right side being the moments of S^{m+2} on the two faces normal to axis i, one dimension down, computed by the same relation. In zero dimensions the moment is the corner value c^{m+2}. Because $t_j = l_j + \frac{1}{2} + \frac{1}{2} x_j$, multiplication by t_j and by t_j^2 is a banded operator on the Legendre index (three-term and five-term), so each relation couples Λ at the indices $k_j - 2$ to $k_j + 2$ in every axis.

The solution. The anchor is taken as the axis of the largest k_i. The relations for $\mathbf{k} \in [0, N)^d$, with $\Lambda = 0$ beyond, form one sparse linear system for the N^d unknowns: the boundary-value formulation of a recurrence whose wanted solution is the minimal one, solved directly rather than marched [13, 19]. The only data are the corner values of S^{m+2} and its analogues one and two dimensions down; no logarithm, inverse hyperbolic sine or arctangent is evaluated, the transcendental functions being carried by the system. Against forty-digit quadrature the edge moments come out to full relative accuracy, the high-order ones included: for $k \leq 16$ the absolute errors relative to $\Lambda[0]$ are 10^{-20} to 10^{-35} while the moments themselves fall to 10^{-13} to 10^{-21} of $\Lambda[0]$, at $\rho = 1$, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{6}$ and 3 and for $m = -3$, -1 and 1. The face moments are accurate to 9×10^{-18} to 3×10^{-16} relative to $\Lambda[0, 0]$, including the faces at $\rho = \sqrt{2}$ and $\rho = 1$ where marching is unstable. The truncation margin used in these tests is 16 Legendre degrees beyond the highest needed; in the coupling blocks 8 suffices (§6).

5.3 Pieces away from the singular point

In $\mathbf{t}$, a piece of W away from the singular point is a box of side 2 in one orthant (each axis interval is $[0, 2]$, $[2, 4]$ or their reflections), or a face of one at $|t_i| = |b - \sigma_i^*|$ for a plane delta. Reflecting each axis to $t_j \geq 0$ and scaling by $\mathbf{t} = 2\tilde{\mathbf{t}}$ makes it the unit box of §5.2, with $c^2 = (t_i/2)^2$ for a plane

delta and $c = 0$ otherwise. On each free axis the weight $w_j(\sigma_j)$ is written as a Legendre series in the box's own coordinate x_j; the kernel monomial $\tilde{t}_j^{\alpha_j}$ multiplies that series in the Legendre basis exactly; and the piece is the contraction

$$\int_{\text{piece}} W\,\mathbf{t}^{\boldsymbol{\alpha}}|\mathbf{t}|^n = 2^{\,n+|\boldsymbol{\alpha}|+d}\prod_j(\pm 1)^{\alpha_j}\sum_{\mathbf{k}}\Big(\prod_j\big[\tilde{t}_j^{\,\alpha_j}w_j\big]_{k_j}\Big)\,\Lambda[\mathbf{k}], \tag{15}$$

with the fixed coordinates' factors (the jump of w' and $t_i^{\alpha_i}$) in front. The Legendre coefficients of the weights and the moments are of order one or smaller, and the moments decay with their degree, so the contraction has no large terms to cancel.

5.4 Pieces at the singular point

A piece whose closure holds the singular point is, reflected and scaled, $[0,1]^d$ with a corner at the singular point ($d = 3$, or $d = 2$ for a plane-delta face through it). Its moments do not exist in the form (12): the moments of S^{-3} alone diverge at the corner, logarithmically in three dimensions, although the integrand the kernel actually presents, $\mathbf{t}^{\boldsymbol{\alpha}}|\mathbf{t}|^{-3}$ with $|\boldsymbol{\alpha}| = 2$, is integrable; the separate moments do not see the factor $\mathbf{t}^{\boldsymbol{\alpha}}$. The piece is split instead into d Duffy pyramids about the singular corner. In pyramid k, $\tilde{t}_k = \rho$ and $\tilde{t}_j = \rho v_j$ for the other axes, $\mathbf{v} \in [0,1]^{d-1}$, so $\mathrm{d}^d\tilde{t} = \rho^{d-1}\mathrm{d}\rho\,\mathrm{d}^{d-1}v$ and $|\tilde{\mathbf{t}}|^n = \rho^n(1+|\mathbf{v}|^2)^{n/2}$, and

$$\int_{[0,1]^d} P(\tilde{\mathbf{t}})\,|\tilde{\mathbf{t}}|^n\,\mathrm{d}^d\tilde{t} = \sum_k \int_{[0,1]^{d-1}} (1+|\mathbf{v}|^2)^{n/2}\,G_k(\mathbf{v})\,\mathrm{d}^{d-1}v, \qquad G_k(\mathbf{v}) = \sum_p \frac{c_p(\mathbf{v})}{p+n+d}, \tag{16}$$

where P is the weight times the kernel monomial and $c_p(\mathbf{v})$ is the coefficient of ρ^p in $P(\rho\,\mathbf{e}_k(\mathbf{v}))$. The radial integral is exact. The coefficients that vanish because W vanishes at the vertex, to order $1 - n_{\text{der}}$ on an axis where the vertex is an outer end of the piece ($\sigma = \pm 2$) with n_{der} derivatives moved onto that axis, and to order zero where it is the inner breakpoint $\sigma = 0$, are set exactly to zero, so that $p + n + d > 0$ wherever $c_p \neq 0$; a non-zero coefficient with $p + n + d \leq 0$ would be a divergent integral and is reported as an error. G_k is a polynomial on the far face of the pyramid, at unit distance from the singular point. It is projected onto Legendre polynomials on $[0,1]^{d-1}$ by a Gauss rule exact for its degree (24 points per axis, 16 coefficients), which here only manipulates polynomials and never integrates the singular kernel, and contracted with the face moments Λ of $(1+|\mathbf{v}|^2)^{n/2}$ from §5.2, with $c^2 = 1$ and $\mathbf{l} = 0$.

6 Verification and accuracy

The closed forms are judged against a reference that shares no step with them, and where no such reference exists, against quadrature that the reference has shown to be accurate.

6.1 A forty-digit reference

A reference must share no step with the routes it judges. For corner and edge contact one exists that is also simple. There the singular point is a vertex of the pieces it touches, and W_{ac}

vanishes there as ρ^3 (corner) or ρ^2 (edge), ρ the distance from the singular point. The most singular static term, $\partial^4 r \sim \rho^{-3}$, is then absolutely integrable, and so is every other: the block is an ordinary integral, with no distribution, no moved derivative and no master integral. The reference evaluates

$$\int_{[-2,2]^3} W_{ac}(\mathbf{s})\, \partial^t r^m(\mathbf{R}+\mathbf{s})\, \mathrm{d}^3 s \tag{17}$$

for each of the 41 static terms ($m = -1$ and $m = 1$) at forty digits, at $h = 1$. The factors of W_{ac} are integrated exactly from their definition. The domain is cut into its 8 pieces; on a piece with the singular point at a vertex, three Duffy pyramids [2] about that vertex make the integrand u_1^k times a smooth function, with $k \geq 0$ an integer, so a tensor Gauss rule converges exponentially; the other pieces are smooth. The terms are assembled into the static block with exact rational weights and the medium's constants in exact arithmetic. Three independent checks establish its accuracy.

(a) A polynomial kernel, $|\mathbf{R}+\mathbf{s}|^2$, for which Mathematica gives the six-dimensional integral of Eq. (1) exactly. The reference reproduces it to 10^{-35}, which checks W, the pieces, the Duffy map and the weights.

(b) Integration by parts on the hardest term: with $F = \partial_0\partial_0\partial_1 r$, $\int W\, \partial_2 F = -\int (\partial_2 W)\, F$, the boundary terms vanishing because W is continuous and zero on the boundary of its support. The two sides have different integrands and agree only if the singular integration is accurate. They agree to 1.4×10^{-24} (corner) and 2.4×10^{-24} (edge) for the degree-one field and 3.9×10^{-24} and 3.5×10^{-24} for the degree-two one.

(c) Convergence: the rule at 20 and at 28 points per axis agree to better than 10^{-23} in every case (3×10^{-25} to 2×10^{-24} where the value was recorded). Each piece converges by about five digits per four points.

Pointwise identities such as $\nabla^2(1/r) = 0$, which hold at every node of the rule, would pass whatever the integration did, and are not used as checks.

6.2 Accuracy of the blocks

Blocks against the reference. With pieces away from the singular point by Eq. (15) and pieces at it by Eq. (16), the static blocks agree with the reference as in Table 2 (relative error in the Frobenius norm).

	degree one		degree two	
	corner	edge	corner	edge
power basis (§4)	2.3×10^{-13}	9.4×10^{-14}	1.7×10^{-11}	5.5×10^{-12}
Legendre basis (§5)	1.0×10^{-15}	9.7×10^{-16}	3.7×10^{-15}	4.3×10^{-15}
quadrature, 20 points	1.1×10^{-15}	1.8×10^{-15}	1.9×10^{-15}	2.7×10^{-15}

Table 2. The static coupling block against the forty-digit reference. A degree-one field has 4 test functions and 10 source monomials, a degree-two one 10 and 35; each block has 81 entries per pair.

The remaining gap to quadrature for the degree-two field, a factor of 1.3 to 1.6, is round-off, not a residual defect. Term by term, quadrature's own worst moments are in error by 2 to 3×10^{-14} relative to the largest entry of the term and the stable evaluation's by about 5×10^{-14}, with a median of about 10^{-15} for both. Forming the weights' power coefficients in exact rational arithmetic before rounding gains only 15 to 20%.

Self and face contact. The reference does not cover self and face contact, where the static terms are distributions. There the stable evaluation is compared with quadrature, which the reference shows to be accurate at corner and edge contact. At $h = 1.25$, for all 41 static terms and all pairs of the degree-one field, the two agree to a worst term of 4.2×10^{-15} (self), 9.9×10^{-15} (face), 3.3×10^{-14} (edge) and 1.8×10^{-14} (corner), with medians of 1.0 to 1.7×10^{-15}. These comparisons are made at $h \neq 1$ on purpose: they also check the physical scaling of each piece by powers of h, which a reference at $h = 1$ cannot see.

The complete blocks. With the dynamic terms of the series (3) added, the whole touching block, static and dynamic, is the series summed term by term, each term a universal moment times $h^{6+m-|\iota|}$ and the weights c_1, c_2. At $k_S h = 0.0625$ it agrees with the quadrature block at 20 points per axis to about 10^{-15} for the degree-one field at all four contact geometries, and to 2 to 4×10^{-15} for the degree-two field; in the power basis the degree-two field's blocks stall at 9×10^{-15} to 2×10^{-11}.

Reciprocity. Two further symmetries hold for every touching block, not only the self block. The propagator satisfies $\mathbf{W}\mathbf{P}(\mathbf{r})$ symmetric at every $\mathbf{r}$, with $\mathbf{W} = \mathrm{diag}(\mathbf{I}_6, \frac{1}{2}\mathbf{I}_3)$ (the factor $\frac{1}{2}$ on the three engineering shears), and $\mathbf{P}(-\mathbf{r}) = \mathbf{\Pi}\mathbf{P}(\mathbf{r})\mathbf{\Pi}$ with $\mathbf{\Pi} = \mathrm{diag}(-\mathbf{I}_3, \mathbf{I}_6)$. Every block between test functions is therefore $\mathbf{W}$-symmetric, $(\mathbf{W}\mathbf{K}_{ab})^{\mathsf{T}} = \mathbf{W}\mathbf{K}_{ab}$, and the self block obeys $\mathbf{K}_{ba}(0) = \mathbf{\Pi}\mathbf{K}_{ab}(0)\mathbf{\Pi}$, by exchanging the two points of the cell. At the self, face, edge and corner geometries, by quadrature and by the series in k of §7.3 alike, the first holds to 3×10^{-21} of the largest entry and the second to 6×10^{-17}.

7 Cells that do not touch, and the frequency sweep

For two cells that do not touch the kernel is smooth over both cells, and the block can be obtained without a singular integral, either by the two-centre series of §3, which now converges, or by quadrature of the s-form. This section compares the routes, and then separates what is computed once from what each frequency costs.

7.1 The two-centre series for elastic waves

The word multipole is used in two senses in what follows, and the difference matters. In the first, a source is resolved into its polynomial moments while the Green's tensor is integrated exactly: the coupling blocks of the cells (§2), which are exact at every separation. In the second, the usual one, the kernel itself is expanded about a centre and truncated, which converges only when

the cells are far enough apart: the series for distant cells of §7.2. The second is what the fast multipole method rests on [20]; four things make it different for elastic waves and polynomial cubes.

Two wave speeds. The Green's tensor (3) is a shear term plus the gradient of a difference of a shear and a compressional term divided by k_S^2 [21], and that difference is finite at small distance only by cancellation. No single addition theorem serves, and the cancellation has to be removed analytically rather than numerically: evaluated as the difference of the two spherical-Hankel forms, the derivatives of $(g_S - g_P)/k_S^2$ lose accuracy as $\varepsilon/(kR)^2$, ε the machine precision. The blocks of cells 3 to 64 cells apart at $k_S h = 10^{-4}$ are then wrong by 3×10^{-9} to 10^{-12}, as that law predicts; with the derivatives taken from the power series of the difference for $k_S r \le 1$, they are accurate to 10^{-15} to 8×10^{-15}.

A nine-component state. The state carries an independent strain, so the propagator $\mathbf{P}$ carries up to two more derivatives than the Green's tensor, and the sources are forces and stresses rather than charges. The derivatives of every order are combinations of the radial functions $F_q = (r^{-1}\mathrm{d}/\mathrm{d}r)^q f$ of two scalars, g_S and $(g_S - g_P)/k_S^2$, through fixed tables.

Cubes carrying polynomials. The natural multipoles are not spherical moments of a point source but Cartesian moments of two cells at once, $\mu_{ac}^{\gamma} = \iint \xi^{\mathbf{e}_a}(\xi - \xi')^{\gamma}\xi'^{\mathbf{e}_c}\mathrm{d}\xi\,\mathrm{d}\xi'$, products of one-dimensional integrals of monomials, rational numbers.

The radius of convergence. The separation $\mathbf{s}$ of two points of the cells ranges over $|\mathbf{s}| \le 2\sqrt{3}\,h$, so the expansion about the centre separation $\mathbf{R}$ converges geometrically in $\rho = 2\sqrt{3}\,h/R$, the extent of both cells over their distance, and the wavenumber enters only through the derivatives at $\mathbf{R}$: the term of order n is bounded by $\sum_j \rho^{n-j}\kappa^j/j!$ with $\kappa = 2\sqrt{3}\,k_S h$. There is no restriction on $k_S R$, but the series needs $\rho < 1$.

7.2 Routes for cells that do not touch

The two-centre series (6) converges when the circumscribing spheres of the two cells do not overlap. It is built from the two cells' moments μ_{ac}^{γ} of §7.1 and the derivatives of the propagator at $\mathbf{R}$ from those of the two radial scalars. The derivatives of g_k are spherical Hankel functions, $F_q = \mathrm{i}(-1)^q k^{q+1} h_q(kr)/r^q$; those of $(g_S - g_P)/k_S^2$ come from its power series for $k_S r \le 1$ and from the difference of the two Hankel forms beyond (§7.1). The series converges geometrically in $\rho = 2\sqrt{3}\,h/R$ and is refused for $\rho > 0.6$, about three cells apart; it is truncated at the order where the bound of §7.1 falls below the tolerance.

Every pair that does not touch. Close to contact the series about the cell centres converges too slowly: at two cells apart $\rho = 0.87$. The cross-correlation W_{ac} is a polynomial on each of the eight boxes of its support, so each box can be expanded about its own centre $\mathbf{s}_B$, the moments being those of a polynomial over a box, exact, and the ratio becoming the box's half-diagonal

over $|\mathbf{R} + \mathbf{s}_B|$: 0.52 at two cells apart, and 0.33 after one bisection of the boxes, each bisection halving it. The order is chosen with the bisection level to minimise the cost, subject to one constraint: at the offset $(2, 1, 1)$ a plan of order 48 to 56 without bisection stalls at 6×10^{-13} to 2×10^{-12}, while one bisection at order 36 to 40 reaches 10^{-15} to 4×10^{-15}. The rounding of a series that long grows with its order, so the order is capped at 40 and a plan that needs more bisects instead.

Comparison. For a degree-one field, at offsets from two to sixty-four cells and at $k_S h = 10^{-4}$, 0.0625, 0.5 and 2, every route was compared with the s-form integrated by Gauss rules of 32 points per axis on each of its eight pieces; that reference agrees with 24 points to 2×10^{-15} (to 10^{-14} at $k_S h = 2$). Four routes were timed per block, warm, the one-off moments and coefficients excluded: the s-form with a Gauss rule, the series (6), its piecewise form, and the frequency-independent series in k of §7.3.

Accuracy. With the two choices below, every route is at round-off wherever it applies: the largest relative error over all offsets and frequencies is 9×10^{-15} for the Gauss rule and 10^{-14} for the two multipole forms, the same as the reference's own agreement between 24 and 32 points. The series in k is at round-off where it converges, and loses accuracy as $k_S r$ grows, 5×10^{-13} at sixty-four cells for $k_S h = 0.0625$, because its terms grow like $(k_S r)^t / t!$ before they fall; beyond about $k_S r = 14$ it refuses. Two choices are needed for this. The derivatives of $(g_S - g_P)/k_S^2$ are taken from its power series for $k_S r \le 1$ (§7.1): as a difference of Hankel forms they cancel, and the multipole blocks at $k_S h = 10^{-4}$ would be wrong by 3×10^{-9} to 10^{-12}, the piecewise form near contact by up to 9×10^{-5}. And the order of the Gauss rule must grow with the frequency: an order set by distance alone, eight points per axis beyond four cells, leaves the blocks wrong by 3×10^{-11} at $k_S h = 2$ and by 10^{-6} at $k_S h = 4$. Against forty points, the order that reaches about 10^{-14} is the larger of the distance rule and $8 + 2\lceil k_S h - 1 \rceil$, for offsets from two to thirty-two cells and $k_S h$ up to 4.

offset	Gauss	multipole series	piecewise series	series in k
$(2, 0, 0)$	28 ms	refused	15.8 s	0.8 ms
$(2, 1, 1)$	26 ms	refused	15.7 s	0.8 ms
$(3, 0, 0)$	13 ms	9.6 s	2.2 s	0.9 ms
$(4, 3, 2)$	12 ms	0.62 s	0.73 s	1.1 ms
$(8, 0, 0)$	8 ms	0.29 s	0.49 s	refused
$(64, 0, 0)$	8 ms	0.12 s	0.38 s	refused

Table 3. Cost per block for a degree-one field at $k_S h = 0.5$, every route accurate to round-off. At $k_S h = 2$ the Gauss rule costs 11 to 32 ms, the two multipole forms 1.5 to 50 s, and the series in k is refused at every offset; at $k_S h = 10^{-4}$ the multipole series falls to 15 ms at sixty-four cells.

What the routes are for. On a regular grid the distant blocks are needed once per offset orbit and frequency, and the Gauss rule of the s-form, with its order now tied to the frequency, is the practical route: it is accurate to round-off at every separation and frequency tested, at 8 to 32 ms a block. The series in k is faster still, under a millisecond a block, wherever $k_S r$ is

small enough for it, which is the touching and near cells of a sweep at low frequency (§7.3). The multipole series is the slowest, by one to four orders of magnitude, and it has one property the others lack: it is exact term by term, with no restriction on $k_S R$, so it is the independent check of the other two, and it is the form a fast multipole method would use for cells far apart (§9).

A degree-two field. The same comparison for a degree-two field (10 test functions, 35 source monomials), at offsets $(2,0,0)$, $(2,1,1)$, $(4,3,2)$ and $(16,0,0)$ and at $k_S h = 10^{-4}$, 0.5 and 2, puts every route at round-off again: the largest error is 2×10^{-14}, the Gauss rule at sixteen cells and $k_S h = 2$. The costs rise with the block: 25 to 93 ms for the Gauss rule, 53 ms to 3.1 s for the multipole series, 1 to 172 s for its piecewise form, and 1.8 to 5.5 ms for the series in k where it converges. The ranking of the routes is unchanged.

7.3 Frequency-independent coefficients, and the frequency sweep

A synthetic seismogram needs the coupling blocks at every frequency of a sweep, at least 256 of them. What can be computed once, for the geometry alone, and what must be computed again at each frequency, decides the cost. The universal moments of the touching blocks are of the first kind. The same structure serves the blocks between cells that do not touch.

The series for cells that do not touch. For two cells that do not touch, the kernel is smooth on every piece of W, and the series (3) applies term by term with coefficients obtained by a Gauss rule on the unit cell, once for each offset:

$$\mathbf{K}_{ac}(\mathbf{R}) = \sum_{t \geq 0} \Big[c_1 (t-1)\, \mathbf{S}^A_{ac}(t;h) + c_2 (t-1)\, \mathbf{S}^B_{ac}(t;h) \Big], \qquad \mathbf{S}(t;h) = h^{\,6+(t-1)-|\mathbf{a}|}\, \mathbf{S}(t;1), \tag{18}$$

where $\mathbf{S}^A$ and $\mathbf{S}^B$ are the s-form integrals (2) of the propagator built from $\delta_{ij} r^{t-1}$ and from $\partial_i \partial_j r^{t-1}$, and $|\mathbf{a}|$ is the number of derivatives on r^{t-1} in each of the 81 entries (two more for the second family). Because quadrature is linear in the kernel, the series summed with these coefficients equals the Gauss block exactly, up to the truncation of the series. It agrees with the Gauss block to 10^{-15} to 10^{-13} for offsets from two to five cells and $k_S h$ from 10^{-6} to 0.5, and costs 0.3 ms per block and frequency against 9 ms for the Gauss block.

The stopping rule. The weight c_2 carries k_S^{-2}: its term t is of order k^{t-2}, as large as the c_1 term $t-2$, and the radiation (imaginary) part at order k comes from $t = 1$ of the first family and $t = 3$ of the second together. Each series is therefore summed through $t = 4$ at least, and until $(k_S r_{\max})^{t-2}/(t-2)!$ falls below the tolerance, $r_{\max}$ the largest separation of two points of the cells. A rule that bounds term t by $(k_S r_{\max})^t/t!$, ignoring the k_S^{-2}, stops two powers early at low frequency: the touching blocks' imaginary part is then wrong by 7×10^{-9} at $k_S h = 10^{-4}$, and the whole block by 10^{-12}, and the imaginary part violates its low-frequency law, $O(k)$ with an $O(k^3)$ correction, which the rule above preserves.

The cost of the touching blocks. The universal moments are computed once for each contact geometry and cell degree: they depend on neither the frequency nor the cell size. For all four geometries and both degrees, with the series carried to the powers that $k_S h = 0.0625$ needs, that one-off step took 3745 s on one machine, against 998 s in the power basis: about 0.75 s per term against 0.3 s, over some 2600 terms. Stability has a price, and it is paid once. Each block thereafter costs the assembly of the series alone, 5 ms for the degree-one field and 33 ms for the degree-two one, against 0.4 to 13 s for a quadrature block at the number of points (14 to 20 per axis) that reaches 10^{-15}. One frequency needs all eight blocks: 17.4 s by quadrature at 14 points (2.3 s for the degree-one field, 15.1 s for the degree-two field), 0.15 s by the series. The one-off cost is therefore recovered after about 220 frequencies, and a synthetic over 256 frequencies is already cheaper by the series. Two choices keep the one-off cost down, by a factor of 2.3, without changing any block beyond round-off: a truncation margin of 8 Legendre degrees (the blocks are the same to the digits shown for margins from 4 to 16), and one truncation shared by every term of a cell degree, so that the sparse systems are solved once and reused.

The sweep. On the grid of the graded sphere of Nestor [1] (n cells across a sphere of radius a), all the blocks of one frequency were formed for the Legendre cell with a field and a contrast of degree one, at 256 frequencies $k_S a = 0.5\, j/256$. The cost per frequency was timed at $j = 1$, 64, 160 and 256, and the total is the one-off cost plus 256 times their mean, on one machine with eight threads.

n	quadrature	series
6	8.5	5.7
10	12.8	7.0
14	22.2	11.9

Table 4. Minutes for the blocks of 256 frequencies, one-off cost included. The series costs 336, 403 and 684 s once (the universal moments of the touching blocks and the coefficients of every other offset) and 0.03, 0.06 and 0.12 s per frequency; quadrature 2.0, 3.0 and 5.2 s per frequency.

Over the sweep the series is the cheaper route by a factor of 1.5 to 1.9, although the stable evaluation of §5 has raised its one-off cost by 14 to 94% (from 173, 302 and 601 s), and it agrees with the quadrature blocks to round-off. The comparison is of the blocks alone: the solve, repeated at each frequency, is not included.

Longer sweeps. With the costs of Table 4, 1024 frequencies cost the Legendre cell 6.1, 7.7 and 13.4 min by its series in the wavenumber, one-off cost included, against 34, 51 and 89 min for quadrature at each frequency ($n = 6$, 10 and 14). The gain is the series'; the closed forms are what make the series exact for the touching blocks, at a one-off cost that over such a sweep does not matter.

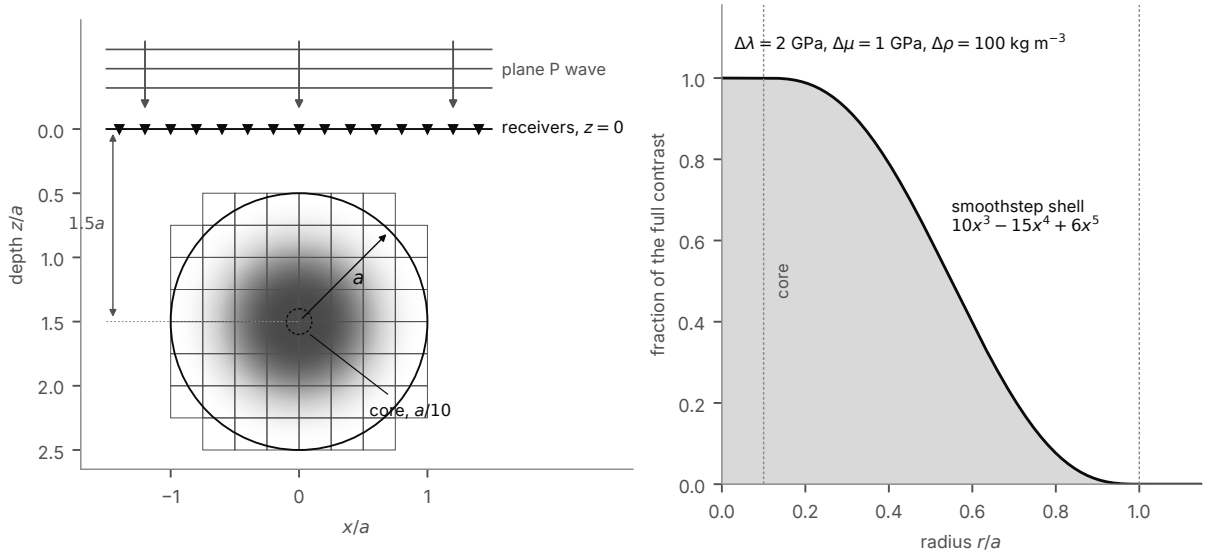


Figure 4. The graded sphere. Left: the section through the centre in the plane of incidence, shaded by the contrast, with the homogeneous core of radius $a/10$ (dashed), the cells of the model of 8 cells across that overlap the sphere, and the plane P wave coming down. The receiver plane and the depth axis are those of the τ–p section of §8.2: receivers at $z = 0$, the centre at depth $1.5a$. Right: the profile, the fraction of the full contrast against the radius, 1 in the core and the smoothstep $10x^3 - 15x^4 + 6x^5$ across the shell, x the fraction of the shell measured from the surface.

8 Applications to a graded sphere

Two applications use what is special about the coefficients: that they are exact, and that they do not depend on the frequency.

8.1 The graded sphere, power by power in the frequency

The frequency-independent coefficients make possible something that a quadrature at each frequency does not: the scattering of a cell model expanded in powers of the frequency, each power obtained on its own and exactly, and compared power by power with the exact body's. The body is the graded sphere of Nestor [1] with its core reduced: radius $a = 10\,\mathrm{m}$, a homogeneous core of radius $a/10$, and a shell across which the contrast ($\Delta\lambda = 2\,\mathrm{GPa}$, $\Delta\mu = 1\,\mathrm{GPa}$, $\Delta\rho = 100\,\mathrm{kg\,m^{-3}}$, in the background $\alpha = 5000\,\mathrm{m\,s^{-1}}$, $\beta = 3000\,\mathrm{m\,s^{-1}}$, $\rho = 2500\,\mathrm{kg\,m^{-3}}$) falls to zero with the smoothstep profile $10x^3 - 15x^4 + 6x^5$, x the fraction of the shell measured from the surface, so that a cell model of it has no staircase. A plane P wave is incident along an axis, and the far field is taken at nine angles in a plane that contains it. The cells are Legendre cells [7], of degree one and of degree two, every cell that overlaps the sphere being kept (Figure 4).

The exact series. The sphere's per-order T-matrix $T_n(w)$, $w = k_P a$, is analytic near $w = 0$. Its Taylor coefficients are obtained by Cauchy's formula on a circle in the complex w-plane, at 90 digits from the radial (Takeuchi–Saito) integration of the graded shell, with no difference quotient in the frequency. Four checks: for the homogeneous sphere the contour coefficients equal its exact symbolic series to every digit; for the graded sphere, contours of radius 0.2 and 0.3 and of 48 and

64 points agree to 8×10^{-41}; the series to w^{24} reproduces the direct T-matrix at $w = 0.01$ and 0.05 to 4×10^{-33}; and as the shell narrows, the coefficients approach the homogeneous sphere's in proportion to its width. The far-field amplitudes follow in closed form,

$$r\,\mathrm{e}^{-\mathrm{i}k_P r}u_r = -\frac{\mathrm{i}}{k_P}\sum_n (2n+1)\,T_n^{PP}(w)\,P_n(\cos\theta), \qquad r\,\mathrm{e}^{-\mathrm{i}k_S r}u_\theta = -\frac{\mathrm{i}}{k_P}\sum_n (2n+1)\,T_n^{SP}(w)\,\frac{\mathrm{d}P_n}{\mathrm{d}\theta}, \quad (19)$$

from $h_n(z)\sim(-\mathrm{i})^{n+1}\mathrm{e}^{\mathrm{i}z}/z$ and the plane wave's $(2n+1)\mathrm{i}^n/(\mathrm{i}k_P)$.

The cell model as a series. Every coupling block is a power series in k_S with the frequency-independent coefficients of §7.3. By Eq. (3) the coefficient of k_S^j is

$$\mathbf{K}_j = \frac{1}{4\pi\mu}\Big[\frac{\mathrm{i}^j}{j!}\,\mathbf{A}\big(r^{j-1}\big) + \frac{\mathrm{i}^{j+2}\big(1-\gamma^{j+2}\big)}{(j+2)!}\,\mathbf{B}\big(r^{j+1}\big)\Big], \qquad \gamma=\frac{\beta}{\alpha}, \quad (20)$$

with $\mathbf{A}(r^m)$ and $\mathbf{B}(r^m)$ the blocks of $\delta_{ij}r^m$ and of $\partial_i\partial_j r^m$: the universal moments for cells that touch and the coefficients of Eq. (18) for the others. The contrast is $\mathbf{\Delta}=\mathbf{\Delta}_0+\beta^2k_S^2\mathbf{\Delta}_1$, the density entering through $\omega^2\Delta\rho$, and the incident wave's Legendre moments are a power series in $k_P=\gamma k_S$. The Galerkin system $(\mathbf{M}-\mathbf{K}\mathbf{\Delta})\psi=\psi^0$ of the cells [7] is therefore a power series in k_S, and so is its solution, $\psi=\sum_j k_S^j\psi_j$, each order one solve with the static operator,

$$(\mathbf{M}-\mathbf{K}_0\mathbf{\Delta}_0)\psi_j = \psi_j^0 + \sum_{i=1}^{j}\mathbf{K}_i\mathbf{\Delta}_0\,\psi_{j-i} + \beta^2\sum_{i=0}^{j-2}\mathbf{K}_i\mathbf{\Delta}_1\,\psi_{j-2-i}. \quad (21)$$

The far field is expanded the same way, through $\mathrm{e}^{-\mathrm{i}k\mathbf{x}\cdot\hat{\mathbf{r}}}$ and the source, so every power of the cell model's amplitude is obtained directly: no frequency is sampled and nothing is fitted. The blocks are applied by fast Fourier transforms on the grid, and since every reflection of the cube changes only the signs of a block's entries, an eighth of the transformed blocks is stored. Two checks. Summed at $k_Sh=10^{-4}$, the $\mathbf{K}_j$ reproduce the series blocks of §7.3 to 2×10^{-16} at the self, face and corner offsets and at two distant ones. And on 4 cells across, the summed series of the far field equals a direct solve at $k_Sa=0.02$ and 0.05 to 4×10^{-12} and 6×10^{-11} of the amplitude.

No quadrature error. Every block in these solves is accurate to about 10^{-14} or better: the touching ones in closed form by the stable evaluation of §5, the others by the Gauss orders of §7. The errors against the exact sphere below are 10^{-4} to 10^{-7} of each coefficient, five or more orders above the agreement of the series with a direct solve: they are the cell model's own, with no quadrature error in them.

What the powers show. With the degree-one cell every coefficient converges at fourth order (Table 5): between 18 and 20 cells across the apparent orders are 3.96 to 4.00, and the cell model reproduces the exact zero of the $(ka)^3$ coefficient to round-off. The orders approach four from below, and not at the same pace: the $(ka)^6$ coefficient is at 4.0 from 6 cells on, while that of $(ka)^7$ starts at 2.7 and reaches 3.97 only on the finest grids. The $(ka)^6$ coefficient carries the largest relative error, 1.4×10^{-1} on 4 cells and 2×10^{-4} on 20, but it is small, four thousandths of

cells across	cells	$(ka)^2$	$(ka)^3$	$(ka)^4$	$(ka)^5$	$(ka)^6$	$(ka)^7$
4	64	2.8×10^{-4}	0	3.2×10^{-4}	2.8×10^{-4}	1.4×10^{-1}	4.2×10^{-5}
6	208	5.9×10^{-5}	0	8.5×10^{-5}	6.0×10^{-5}	2.6×10^{-2}	5.7×10^{-5}
8	408	2.0×10^{-5}	0	3.0×10^{-5}	2.1×10^{-5}	8.1×10^{-3}	2.7×10^{-5}
10	768	8.7×10^{-6}	0	1.3×10^{-5}	8.9×10^{-6}	3.3×10^{-3}	1.2×10^{-5}
12	1280	4.3×10^{-6}	0	6.4×10^{-6}	4.4×10^{-6}	1.6×10^{-3}	6.2×10^{-6}
14	1912	2.3×10^{-6}	0	3.5×10^{-6}	2.4×10^{-6}	8.6×10^{-4}	3.4×10^{-6}
16	2776	1.4×10^{-6}	0	2.1×10^{-6}	1.4×10^{-6}	5.0×10^{-4}	2.1×10^{-6}
18	3888	8.7×10^{-7}	0	1.3×10^{-6}	8.8×10^{-7}	3.2×10^{-4}	1.3×10^{-6}
20	5256	5.7×10^{-7}	0	8.6×10^{-7}	5.8×10^{-7}	2.1×10^{-4}	8.6×10^{-7}
order $6 \to 8$		3.70		3.60	3.70	4.04	2.67
order $12 \to 14$		3.92		3.90	3.93	4.01	3.81
order $18 \to 20$		3.96		3.97	3.96	4.00	3.97
size		1	0	8.8×10^{-2}	1.9×10^{-3}	4.0×10^{-3}	3.9×10^{-4}

Table 5. The graded sphere's far-field amplitude, power by power in $k_S a$: the relative error of each coefficient of the degree-one Legendre cell model against the exact series, the largest over the P and S amplitudes and the nine angles. The $(ka)^3$ coefficient is zero for the exact sphere and for the cell model alike, to 2×10^{-16} of the leading term. The last row is each coefficient's size against the leading one.

the leading term, so in a solve at one frequency it is hidden in the total. A solve at one frequency sees only the sum; here each power is measured on its own.

Cells of degree two. With field and contrast of degree two in each cell, every coefficient is an order of magnitude or more smaller on every grid, and the apparent orders run above four (Table 6): between 10 and 12 cells across they are 6.2 to 7.5, not yet settled at the order $\min(2p + 2, 2r + 2) = 6$ expected of a cell of degree two [1], which finer grids would show. At 12 cells the degree-two cell is 43 to 54 times more accurate than the degree-one cell for every coefficient but $(ka)^6$, and 8100 times for that one. It is also the better cell per unknown: on 10 cells across, with 69,120 unknowns, its leading coefficient is accurate to 2.4×10^{-7}, where the degree-one cell needs 20 cells and 189,216 unknowns for 5.7×10^{-7}. The degree-two blocks cost more once: their near coefficients took 66 min, against three for degree one, and are then reused for every grid and frequency.

cells across	cells	$(ka)^2$	$(ka)^3$	$(ka)^4$	$(ka)^5$	$(ka)^6$	$(ka)^7$
4	64	2.9×10^{-5}	0	4.2×10^{-5}	3.2×10^{-5}	4.3×10^{-4}	3.3×10^{-5}
6	208	3.6×10^{-6}	0	4.3×10^{-6}	3.5×10^{-6}	3.4×10^{-5}	4.8×10^{-6}
8	408	7.9×10^{-7}	0	1.4×10^{-6}	8.1×10^{-7}	4.1×10^{-6}	1.6×10^{-6}
10	768	2.4×10^{-7}	0	4.3×10^{-7}	2.9×10^{-7}	7.7×10^{-7}	5.2×10^{-7}
12	1280	7.9×10^{-8}	0	1.2×10^{-7}	8.3×10^{-8}	2.0×10^{-7}	1.4×10^{-7}
order $6 \to 8$		5.26		3.82	5.15	7.33	3.72
order $10 \to 12$		6.16		6.86	6.82	7.50	7.07

Table 6. As Table 5, for the cell model of Legendre cells of degree two (field and contrast in the ten polynomials of degree at most two, 90 unknowns per cell). The $(ka)^3$ coefficient is again zero to 3×10^{-16} of the leading term.

What the series gives that a solve does not. Each power of the cell model's response is available on its own, exactly, and can be set against the same power of the exact body. Whether a coefficient converges more slowly than the rest, and whether the degree of the cell cures it, are questions that a solve at one frequency, which sees only their sum, cannot ask. They can be asked here because the coupling coefficients are exact and independent of the frequency.

8.2 A τ–p section of the graded sphere

The coefficients give the blocks at any frequency for the price of a weighted sum, so a broadband response costs one solve per frequency. We use this to compute a seismogram with no far-field approximation: the τ–p section of the graded sphere of §8.1, recorded on a plane close to the body, near field and evanescent waves included, from cells of degree one and of degree two, beside the exact sphere's. It shows the coupling integrals of this paper at work in the time domain. The geometry follows seismic practice (Figure 4): z is depth, the receivers lie on the plane $z = 0$, the sphere's centre is at depth $z_c = 1.5a$, and a plane P wave travels down onto it. The section is built in four steps.

1. The plane-wave spectrum, and the exact reference. Above the body the scattered displacement is a superposition of upgoing plane waves (the Weyl representation),

$$\mathbf{u}(\mathbf{x}) = \int \mathrm{d}^2k \left[c_P(\mathbf{k})\, \mathbf{e}_P\, \mathrm{e}^{\mathrm{i}(\mathbf{k}\cdot\boldsymbol{\rho} + k_z^P (z_c - z))} + c_{SV}(\mathbf{k})\, \mathbf{e}_{SV}\, \mathrm{e}^{\mathrm{i}(\mathbf{k}\cdot\boldsymbol{\rho} + k_z^S (z_c - z))} \right], \qquad k_z^{P,S} = \left(k_{P,S}^2 - |\mathbf{k}|^2\right)^{1/2},\ \operatorname{Im} k_z \geq 0, \tag{22}$$

with $\mathbf{k}$ the horizontal wavenumber and $\boldsymbol{\rho}$ the horizontal position. Where $|\mathbf{k}|$ exceeds k_P or k_S the wave is evanescent. Eq. (22) is exact on every plane above the body, at any distance from it, so no far field is needed. For the exact sphere, c_P and c_{SV} follow at every frequency from the coefficients of its outgoing P and S potentials (the graded sphere's series solution of §8.1), each multipole transformed to plane waves in closed form. They are obtained for horizontal slownesses $p = |\mathbf{k}|/\omega$ on both sides of $1/\alpha$ and $1/\beta$; the identity $\sin\theta = |\mathbf{k}|/k$ holds on either branch, so no complex root is taken.

2. The cell model. The Weyl identity $\mathrm{e}^{\mathrm{i}kr}/r = (\mathrm{i}/2\pi) \int \mathrm{d}^2k\, \mathrm{e}^{\mathrm{i}(\mathbf{k}\cdot\boldsymbol{\rho} + k_z|z|)}/k_z$, applied to the Green's tensor, gives the amplitudes of any bounded source as

$$c_P(\mathbf{k}) = \frac{\mathrm{i}}{2\pi k_z^P}\, \mathscr{A}_P(\hat{\mathbf{k}}_P), \qquad c_{SV}(\mathbf{k}) = \frac{\mathrm{i}}{2\pi k_z^S}\, \mathscr{A}_{SV}(\hat{\mathbf{k}}_S), \qquad \hat{\mathbf{k}}_{P,S} = \frac{\mathbf{k} - k_z^{P,S}\, \hat{\mathbf{z}}}{k_{P,S}}, \tag{23}$$

where $\hat{\mathbf{z}}$ points down, so that $\hat{\mathbf{k}}_{P,S}$ is the direction of the scattered wave, travelling up to the receivers. $\mathscr{A}_P$ and $\mathscr{A}_{SV}$ are the radiation patterns of the P and SV waves: along a real unit vector $\hat{\mathbf{r}}$,

$$\mathscr{A}_P(\hat{\mathbf{r}}) = \lim_{r\to\infty} r\, \mathrm{e}^{-\mathrm{i}k_P r}\, \hat{\mathbf{r}} \cdot \mathbf{u}, \qquad \mathscr{A}_{SV}(\hat{\mathbf{r}}) = \lim_{r\to\infty} r\, \mathrm{e}^{-\mathrm{i}k_S r}\, \mathbf{e}_{SV} \cdot \mathbf{u},$$

with $\mathbf{e}_{SV}$ the SV polarisation, perpendicular to $\hat{\mathbf{r}}$ in the plane of incidence (Eq. (25)): each wave projected on its own polarisation, since the two carry different phases. For a propagating wave

$\hat{\mathbf{k}}$ is a real unit vector and $\mathscr{A}(\hat{\mathbf{k}})$ is the far-field amplitude in that direction. For an evanescent wave k_z is imaginary and $\hat{\mathbf{k}}$ is complex. It is then not a unit vector, $|\mathbf{k}|^2 + |k_z|^2 > k^2$, and no limit $r \to \infty$ defines $\mathscr{A}$ there; only $\hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = (|\mathbf{k}|^2 + k_z^2)/k^2 = 1$ survives. The pattern is the analytic continuation from real directions. For a bounded source it is an entire function of the components of $\hat{\mathbf{k}}$, so the continuation is unique, and it is the same expression evaluated at the complex vector. The cells take it directly. Their pattern is a sum over cells of the moments of each cell's polynomial source $\mathbf{s}$ against the plane wave,

$$\int_{\text{cell}} \mathbf{s}(\mathbf{y})\, \mathrm{e}^{-\mathrm{i}k\hat{\mathbf{k}}\cdot\mathbf{y}}\, \mathrm{d}^3y = \mathrm{e}^{-\mathrm{i}k\hat{\mathbf{k}}\cdot\mathbf{y}_c}\, h^3 \sum_c \mathbf{s}_c \prod_{i=1}^{3} F_{e_{ci}}(-k\hat{k}_i h), \qquad F_n(q) = \int_{-1}^{1} \xi^n \mathrm{e}^{\mathrm{i}q\xi}\, \mathrm{d}\xi = \sum_{l \le n} a_{nl}\, 2\, \mathrm{i}^l j_l(q), \tag{24}$$

with $\mathbf{y}_c$ the cell's centre, $\mathbf{s}_c$ the coefficients of its monomials $\xi^{\mathbf{e}_c}$, and a_{nl} the Legendre coefficients of ξ^n. The master functions are the sinc, $j_0(q) = \sin q/q$, and its relatives j_l, which hold for complex q, and are evaluated as spherical Bessel functions rather than in elementary form, which cancels as $q \to 0$. A far-field routine therefore needs only one change, a complex unit direction. The closed form agrees with a 40-digit quadrature of F_n to 2×10^{-15}, and on a solved cell model with a ten-point Gauss rule per cell to 3×10^{-15}.

3. The seismogram. At horizontal slowness p in the plane of incidence, $\mathbf{k} = \omega p\, \hat{\mathbf{x}}$, the slant stack of the displacement on the receiver plane is the spectrum itself, carried up from the centre:

$$\mathbf{u}(\tau, p) = \frac{(2\pi)^2}{\pi}\, \mathrm{Re} \int_0^\infty \mathrm{d}\omega\, W(\omega) \Big[c_P\, \mathbf{e}_P\, \mathrm{e}^{\mathrm{i}k_z^P z_c} + c_{SV}\, \mathbf{e}_{SV}\, \mathrm{e}^{\mathrm{i}k_z^S z_c} \Big] \mathrm{e}^{\mathrm{i}\omega z_c/\alpha}\, \mathrm{e}^{-\mathrm{i}\omega\tau}, \tag{25}$$

with $\mathbf{e}_P = (\hat{\mathbf{x}} p\omega - \hat{\mathbf{z}} k_z^P)/k_P$ and $\mathbf{e}_{SV} = -(\hat{\mathbf{x}} k_z^S + \hat{\mathbf{z}} p\omega)/k_S$ and $W(\omega) = (\omega/\omega_0)^2 \mathrm{e}^{-(\omega/\omega_0)^2} \mathrm{e}^{\mathrm{i}\omega t_d}$ a Ricker wavelet, delayed by $t_d = 6.3/\omega_0 = 21$ ms so that it is causal: before $\tau = 0$ it stays below 10^{-3} of its peak. The factor $\mathrm{e}^{\mathrm{i}\omega z_c/\alpha}$ counts time from the incident wave's crossing of the receiver plane, so τ is a two-way time and reads as depth: a point at the centre would return P peaking at $\tau = t_d + z_c\big(1/\alpha + \eta_P\big)$ and converted S at $t_d + z_c\big(1/\alpha + \eta_S\big)$, with $\eta = (1/c^2 - p^2)^{1/2}$ the vertical slowness. Three bands result: $p < 1/\alpha$, where P and S both propagate; $1/\alpha < p < 1/\beta$, where only S does; and $p > 1/\beta$, where both are evanescent. An evanescent wave decays as $\mathrm{e}^{-\omega(p^2 - 1/c^2)^{1/2} z_c}$ on its way up, so how much of it survives depends on how close the receivers are: on a plane in the far field none does. Beyond $p = 1/\alpha$ the section is not causal in τ for any wavelet: the decay $\mathrm{e}^{-\omega\eta z_c}$ with η real, and the $1/k_z$ of Eq. (23) with k_z imaginary, are filters of zero and of quarter-cycle phase, which spread an arrival to both sides. Here the section before $\tau = 0$ is below 6×10^{-4} of its peak for $p < 1/\alpha$ and at most 8×10^{-3} beyond, and is not shown. The wavelet is centred at $k_S a = \omega_0 a/\beta = 1$ (48 Hz), and the integral sampled at 128 frequencies up to $k_S a = 3$ (143 Hz); the section has 60 slownesses up to $1.5/\beta$, centred in equal intervals so that none falls on a branch point, where $k_z = 0$ and the $1/k_z$ of Eq. (23) is singular.

4. The frequency sweep. Each frequency is one solve. The cell model is 8 Legendre cells across the sphere, 408 cells, of degree one (14,688 unknowns) or degree two (36,720), with the profile and contrast of §8.1. For cells of degree one every block at every frequency is summed

from the stored coefficients: the touching ones from the universal moments, the others from those of Eq. (18), the Gauss block standing in only where the far series would need more terms than it holds. For cells of degree two the far blocks are summed the same way, and the four touching orbits are integrated at each frequency: the universal moments to the order that $k_S a = 3$ needs would cost hours once, the integration of those four blocks 13 s per frequency. The system is solved by GMRES to 10^{-11}, with the blocks applied by fast Fourier transforms (§8.1). For degree one the sweep took 37 min, the exact sphere included; for degree two, at 32 of the same frequencies (every fourth, which the section's 60 ms of signal allows, its period in time being then 0.22 s), it took 24 min, 5 min of it once for the far coefficients and 37 s per frequency after that.

cells		P and S propagate $p < 1/\alpha$	S only $1/\alpha < p < 1/\beta$	evanescent $p > 1/\beta$
degree one	$u_z(\tau, p)$	1.9×10^{-5}	1.2×10^{-4}	6.0×10^{-5}
	$u_x(\tau, p)$	8.9×10^{-5}	6.6×10^{-5}	2.2×10^{-5}
degree two	$u_z(\tau, p)$	2.7×10^{-7}	5.4×10^{-7}	4.2×10^{-7}
	$u_x(\tau, p)$	1.7×10^{-6}	1.7×10^{-6}	1.5×10^{-7}

Table 7. The τ–p section of the cell model, 8 cells across, against the exact sphere (Figure 5): the largest difference over τ and over the slownesses of each band, relative to the peak of the exact section of the same component. The cells of degree two were run at 32 of the 128 frequencies, and are measured against the exact section on the same frequencies.

What the section shows. The diffraction lies on the two-way times of P and converted S from the centre, and its energy gathers at the branch points, where the $1/k_z$ of Eq. (23) is large. The cells of degree one reproduce it: the two sections are indistinguishable at the scale of the plot (Figure 5), and their difference is at most 1.2×10^{-4} of the peak in every band (Table 7), the evanescent band included. In the spectrum, the error of c_P in the propagating band is 4.5×10^{-5} of its largest value from the lowest frequency to $k_S a \approx 2$, and rises to 4.6×10^{-4} at $k_S a = 3$, where a cell is an eighth of an S wavelength across; the wavelet's taper keeps that upper band small in the section. The sphere has no staircase in this cell model (§8.1): the cells carry the smooth profile as a polynomial, and the error is that of the cells' polynomials, with no quadrature error in the blocks. The evanescent band, which only a plane in the near field records, tests the cells' sources and the coupling blocks at every frequency of the sweep, all of them from the same coefficients. The cells of degree two do far better. Their section differs from the exact one by 1.5×10^{-7} to 1.7×10^{-6} of the peak, 40 to 220 times less than the cells of degree one, band by band. Their error in c_P is 1.3×10^{-6} to 1.8×10^{-6} up to $k_S a \approx 2.3$ and 2.6×10^{-6} at $k_S a = 3$: flat across the band, where that of the cells of degree one grows tenfold.

9 Discussion

The closed forms of the touching blocks are exact: evaluated at forty digits they agree with the reference. Their accuracy in double precision is set by the basis in which the polynomial weight meets the kernel, and the classical results on moments [9] and on minimal solutions of recurrences [12, 13] say which basis and how to obtain it: use moments of orthogonal polynomials,

and obtain them from their recurrence posed as a boundary-value problem rather than marched. The relation (13) makes this possible for a power of the distance over a box in any dimension, and the pyramids of §5.4 reduce the pieces at the singular point to faces where it applies. Two consequences are worth stating. The evaluation uses no transcendental function, only corner values and a sparse solve, so it is as easy to carry to other kernels homogeneous about a point as the closed forms were. And the accuracy is now that of quadrature, so the closed forms can be the default for the touching blocks, with their one advantage over quadrature intact: coefficients computed once, for every frequency.

The boundary-element literature reached a different conclusion for the analytic integration of layer potentials, switching to quadrature away from the singularity [10, 11]. For the integrals of this paper that switch is not enough at degree two (§4). Whether the Legendre relation would serve those integrals too we have not examined. We have not found the relation (13), or the use of Legendre moments by a truncated boundary-value problem for integrals over boxes, in the literature we searched; the search was limited, and this is a statement of what we did not find.

Closed forms or quadrature. Coefficients computed by Gauss rules are as independent of the frequency as the closed forms, and over a sweep they cost as little per frequency, so the series in the wavenumber does not by itself justify the closed forms. Four properties do.

- *Quadrature fails silently outside its calibration.* A Gauss rule is accurate where its order was chosen by measurement, and nothing warns when it is used outside that range. Two cases show it. A Gauss order for the blocks of distant cells set by distance alone leaves them wrong by 3×10^{-11} at $k_S h = 2$ and by 10^{-6} at $k_S h = 4$ (§7.2). And for the face block of the static strain coupling of two uniform cells, a tensor Gauss rule whose two cells share one set of lateral nodes moved monotonically through 8.45, 9.20, 9.59, 9.85 and 10.02×10^{-12} as its order rose from 4 to 12, with steps that shrank as if it were converging, towards about 1.05×10^{-11}; the exact value is 3.009×10^{-12}, obtained by a subdivision fixed point and by a reduction to smooth integrals that agree to 10^{-13}. Refinement does not reveal such a limit; an independent exact value does.

- *Each coefficient carries its own error estimate.* A closed form is a sum whose terms are known, so the magnitude M on which its rounding acts can be carried through every step of the evaluation, as in a running error analysis [18], and εM is then an estimate of the error of each coefficient, and $M/|U|$ its condition number (7). Quadrature gives no such estimate without a convergence study, and the last case above is one in which a convergence study misleads.

- *An independent check.* The closed forms share no step with quadrature, so the agreement of the two checks each, and exposes errors that either alone would hide: the cancellation of the two radial scalars in the multipole blocks at low frequency, up to 9×10^{-5} (§7.1, §7.2); a Gauss order that ignores the frequency, above; and, in the other direction, the four to five digits that the closed forms lose in the power basis (§4).

- *Universal tables.* The coefficients depend on neither the frequency, the cell size nor the medium. Computed in closed form and checked once, they can be tabulated and reused

for every model, with no rule to choose and no calibration to trust for each new one. The one-off cost, 6 to 11 minutes for every offset of the grids of Table 4, is paid once for all models and all frequencies.

Relation to the fast multipole method. The fast multipole method and this paper answer different questions. The method accelerates the application of the whole operator to a chosen tolerance without forming its entries: it groups sources in a hierarchy of boxes, represents each group by a truncated multipole expansion, converts it to a local (Taylor) expansion about a distant box, and evaluates interactions by expansions only between sets that are well separated [20]. Its elastodynamic form writes the Green's tensor through an addition theorem truncated at an order adjusted to the level of the tree, uses the expansions only between non-adjacent cells, and leaves adjacent cells to direct boundary-element integration [22, 23]. This paper computes the entries themselves, and the entries it is most concerned with, those of touching cells, are exactly the ones a fast multipole method leaves to direct integration. The two are therefore complementary: on an irregular or very large body the closed forms of §5 could supply the near field of such a method. On a regular grid of cubes the operator is a convolution, and the fast Fourier transform applies it in $O(N \log N)$ operations, so the expansions of §7.1 are needed for exact entries, not for speed. A cube divides into eight cubes, so cubic cells are the leaves of an octree, the tree of the fast multipole method in three dimensions; if cells are refined where the medium varies fastest, the tree becomes the natural structure, and its coupling blocks are those of this paper. That is an outlook, not a result.

Data availability

The Mathematica notebooks and Python scripts that derive and check every result, and the package they test, are available at https://github.com/tmnestor/MultipleScatteringCalculations: the reference `Mathematica/GradedVoxel_CornerReference.wl` and its results, the comparison `scripts/gate_galerkin_corner_reference.py`, the moments `cubic_scattering/graded_voxel/legendre_moments.py`, the blocks `cubic_scattering/graded_voxel/blocks.py`, and the tests `cubic_scattering/tests/test_legendre_moments.py` and `cubic_scattering/tests/test_graded_voxel_closed_reference.py`; the near blocks and their cost, `scripts/measure_galerkin_near_blocks.py`; the two-centre series at contact, `scripts/measure_touching_multipole_series.py`; a face block checked by an exact reduction, `scripts/face_s_rederivation.py`; the distant blocks, `cubic_scattering/graded_voxel/multipole.py` and `scripts/measure_distant_cell_blocks.py`; the sweep, `scripts/measure_frequency_sweep_cost.py`; and the graded sphere power by power, `scripts/measure_graded_sphere_legendre_series.py`, with the exact series `Mathematica/GradedSphere_LowFrequency.wl` and the outputs of the runs reported in `scratch/legendre_sphere_series/`; the diagram of the sphere, `scripts/plot_graded_sphere_diagram.py`; and the τ–p section, `scripts/taup_graded_sphere.py` and `scripts/plot_taup_paper.py`, with the closed-form moments `cubic_scattering/graded_voxel/farfield.py` and the exact plane-wave

spectrum `scripts/gate_sphere_plane_wave_spectrum.py` and the outputs in `scratch/taup/`.

Acknowledgments

This work began thirty years ago in the author's doctoral research at the Research School of Earth Sciences of the Australian National University, Canberra [24].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Software development. Software development followed a human-directed, AI-assisted workflow, organised as a Test Driven Hypothesis Acceptance cycle: each derivation, implementation or numerical claim was stated as a hypothesis together with a test designed to refute it (an independent route, an exact limit, a high-precision reference or a convergence study), and was accepted only when the test passed; a failed test returned the hypothesis for diagnosis, never the test for adjustment. The author designed the software originally, but only directed and validated recent improvements, which build on the author's original Fortran implementation (the `PhD_fortran_code` folder of the repository given under Data availability) and an extensive library of Mathematica notebooks, developed and extended since 1992, that carry out and verify the symbolic derivations; an agentic coding assistant (Anthropic Claude) operated under human-in-the-loop supervision to implement specified changes and diagnostic tests. The design, the choice of numerical methods, and the resolution of the principal technical difficulties were the author's. All AI-generated changes were subject to code review by the author and validated by automated tests and independent checks: the forty-digit reference, reciprocity, analytic limits and convergence.

Manuscript preparation. During the preparation of this work, the author used Claude (Anthropic) in order to search for relevant open-access research, to write the TikZ code of the diagrams and to help redraft the paper into more coherent sections for readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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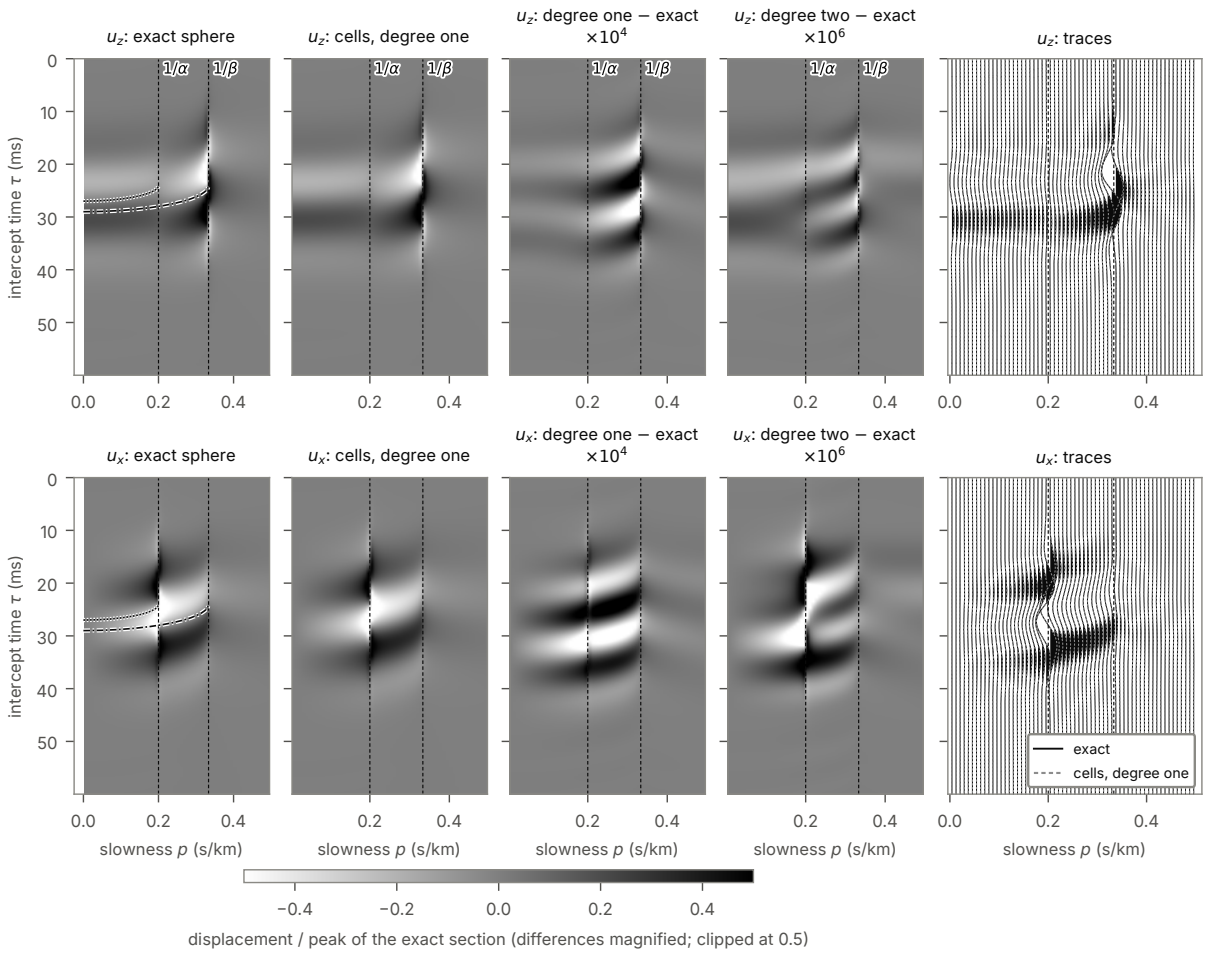


Figure 5. The τ–p section of the graded sphere (Figure 4): plane P wave incident from above, receivers on the plane $z = 0$, the centre at depth $1.5a$; vertical (u_z, top) and horizontal (u_x, bottom) displacement. From the left: the exact sphere; the cell model of 8 Legendre cells of degree one across; the difference of the cells of degree one and of degree two from the exact section, each magnified as marked; and the 60 traces of the section, one per slowness (exact in black with positive lobes filled, cells of degree one dashed over them). Variable density on one grey scale per row, white negative and black positive, the displacement divided by the peak of the exact section and clipped at half of it. Time runs from the incident wave's crossing of the receiver plane, with the wavelet delayed by $t_d = 21$ ms to be causal. The dashed verticals are the branch points $p = 1/\alpha$ and $1/\beta$; the curves on the exact panels are the two-way times of P (dotted) and converted S (dash-dot) from the centre, $t_d + z_c(1/\alpha + \eta)$.